

Supplementary Material

A Promised Grant and Stated Attitudes toward Early Marriage: Experimental Evidence from Burkina Faso

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S1 Design, registration, and measurement

This supplement documents the announcement comparison, selection and measurement checks, and qualifications to the exploratory heterogeneity evidence. T1 offers safe spaces; T2 adds business and vocational training; T3 adds an announced \$400 grant. Every arm, including control, received community sensitization. Outcomes precede payment. References to main-paper displays use descriptive titles; S-prefixed references refer to this supplement.

Registration occurred on 12 October 2020 (RIDIE-STUDY-ID-5f8441974e6ed), after delivery but before follow-up. It specified domains and design features rather than an estimation or multiple-testing plan.

Table S1: Registration-to-analysis crosswalk

Design and analysis	Registration status
Village randomization; arms	Registered; 200 villages planned, 173 at baseline implementation, 162 at follow-up.
Outcome domains	Registered: early marriage/childbearing, economic activity, empowerment, SRH knowledge, and life skills. Indices and testing families defined later.
Announcement contrast/index	Post-registration T3–T2 comparison; index constructed after component results were known.
Participation/head outcomes	Post-registration analyses. Participation is chosen and responds to assignment.
Eligibility, item, selection audits	Added at the analysis stage.
Mental health	Not named in registration; follow-up-only module. Alternative scores/self-worth analyses are post-registration.
Village moderators	Norms, SRH access, market/road access are registered dimensions; indices/tertiles are later choices. Security and exogamy are unregistered.
Estimation and heterogeneity	ANCOVA/PDS-LASSO, RI/Holm, GML, participation decompositions, and theory are post-registration. Focal interaction family is exploratory.
Age-gradient hypothesis	Registered but not separately reported; age enters controls and prediction models.

RI = randomization inference; GML = generic machine learning. The registered midline is the follow-up used here; no post-disbursement survey is available.

Sample flow and analysis populations

Of 200 selected villages, 25 could not be reached at baseline because of insecurity and two had fewer than ten eligible girls. The baseline survey interviewed 8,327 girls and young women in 7,241 households. The analysis roster excludes 608 respondents outside its age/marital-status bounds, leaving 7,719 girls in 173 villages. Eleven further villages could not be reached at follow-up.

Table S2: Baseline roster and follow-up coverage

	Control	T1	T2	T3
Baseline girls	1,704	1,881	1,946	2,188
Girls followed up	1,077	1,108	1,232	1,519
Follow-up share (%)	63.2	58.9	63.3	69.4

Total: 4,936 girls in 162 villages, 64.0% of the baseline roster. Counts reproduce the original participant-flow display; outcome-specific samples can be smaller.

Eligibility versus analysis roster. Program eligibility covered out-of-school unmarried girls aged 10–19 and married girls and young women under 24 (including younger married girls). The analysis roster retains unmarried respondents aged 12–24 and married respondents aged 24 or younger. These populations differ; the 608 exclusions are analysis-roster exclusions, not a restatement of program eligibility.

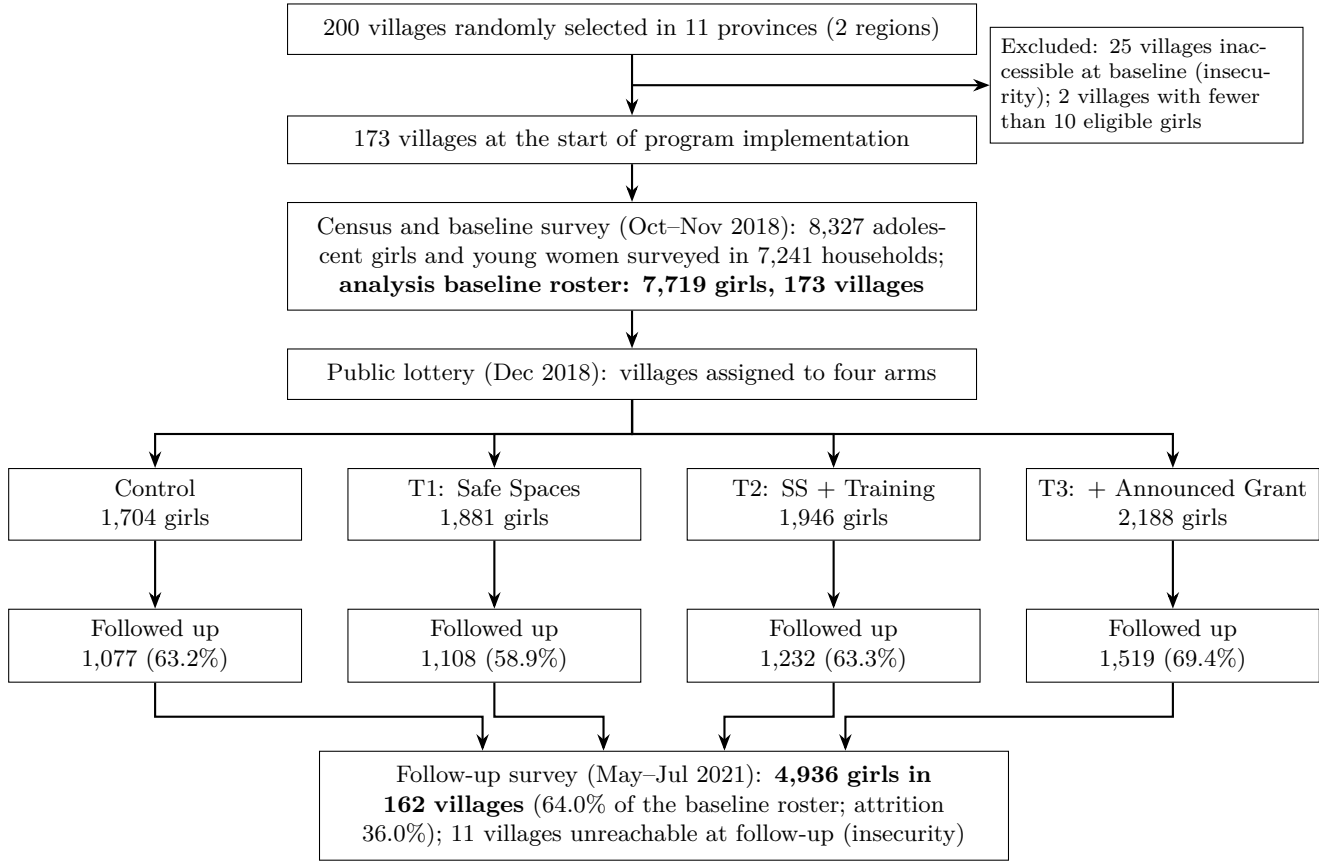
Timing. Baseline: October–November 2018. Public village lottery: December 2018. Safe-space activities: June–December 2019; entrepreneurship training: October–December 2019. Follow-up: May–July 2021. The promised grant was paid subsequently.

Follow-up loss. Household nonresponse accounts for 14.0% of the baseline roster; another 22.4% comprises girls whose households were reached but who were not re-interviewed. Among 758 exited girls with a head-reported marital status, all were recorded as married. Head outcomes cover 6,642 girls (86% of the roster), including 1,727 girls not themselves re-interviewed, and 5,875 distinct heads. Coverage does not differ detectably by arm in that module.

Conditional outcomes. Age at first marriage is observed only for girls married by follow-up; couple decision-making only for girls in a union. These comparisons condition on potentially treatment-affected status and are descriptive. Pregnancy before 18 is not conditioned on marriage, but girls still younger than 18 who have never been pregnant are right-censored and left missing. Other outcomes can have item-specific missingness. Bound tables state their selection target.

Figure S1 shows the participant flow from village selection through the baseline roster, assignment, and follow-up, under the denominator convention stated above: all attrition percentages are relative to the baseline analysis roster.

Figure S1: Participant flow



Notes: All percentages use the analysis baseline roster of 7,719 girls (or its arm-specific counts) as the denominator; on this convention overall attrition is 36.0 percent. The 8,327 girls surveyed at the census stage include 608 respondents outside the study’s eligibility rules (unmarried girls outside ages 12–24 and married respondents older than 24), who are excluded from the analysis roster; relative to the larger surveyed count, follow-up coverage is 59.3 percent, which is the source of the alternative “40.7 percent” attrition figure. Village flow: 200 selected, 173 at implementation, 162 observed at both survey rounds.

Outcome definitions and coding

Table S3: Outcome measures used in the retained evidence

Measure	Definition and interpretation
Early-marriage attitudes	Any perceived disadvantage (“Dis M. Early Any”) and number of disadvantages (“Nb”); any perceived advantage (“Adv M. Early Any”) and number of advantages. Follow-up codes seven disadvantage and three advantage items. Baseline records any mention, not comparable counts.
Announcement index	Four early-marriage attitude outcomes oriented against early marriage, in control-SD units; higher means less favorable attitudes toward early marriage. Post-hoc construction and T3–T2 contrast.
Marriage and fertility	Ever married, marriage before 18, ever pregnant, current pregnancy, pregnancy before 18, and age at first marriage. Binary ITTs use original adverse coding. See the preceding page for conditional outcomes and censoring.
Emotional violence	Any of three emotional-violence experiences in the preceding 12 months. Positive main-ITT coefficients are adverse. GML uses the favorable complement when labeled “No Emo. Viol.”
Income generation (IGA)	Anderson index of paid activity in the previous 30 days, cash profits (inverse hyperbolic sine), own-initiative activities, and number of activities in the previous six months (Table S37).
Mental health	Eight GHQ-style items, four response categories, oriented through response wording so higher is better; no additional subset reversal. Control-normalized, measured at follow-up only. Alternatives appear in Table S35.
Participation	Any self-reported safe-space attendance. No session count is available. Training participation is separate.
Other indices	Finance: borrowing, saving, amount saved. SRH knowledge: contraception/HIV knowledge. SRH behavior: sexual activity/contraception. Menstrual hygiene: knowledge, acceptability, use of modern products. Further indices cover gender attitudes, aspirations, decision-making, socio-emotional skills. SRH and menstrual-hygiene measures apply to baseline ages 15 or older.

Anderson indices use inverse-covariance weights on oriented components. Tables retain reported units and samples. “Any” is an indicator; “Nb” a count.

Signs. Heterogeneity tables often use favorable complements (never married, never pregnant, not currently pregnant). Pregnancy before 18 is not inverted. Read each table’s coding. More disadvantages and fewer advantages both mean attitudes less favorable to early marriage.

Baseline village characteristics

Table S4: Construction of village-level indices

Index	Components and direction
Unrestricted (norms)	Four household restrictions on marrying into/out of another ethnicity or religion (reversed); no cost of renegeing on a marriage promise; village/ethnic-group ideal marriage ages; girl's final say; head's autonomy score; head's minimum marriage age. Higher = more permissive.
Birthcontrols (SRH access)	Distances to condoms, other contraceptives, health center (reversed); public-health/family-planning agent visits in the preceding year. Higher = greater access.
Public infrastructure	Distances to identity/birth-registration services, year-round/seasonal roads, bush-taxi stations, elementary/secondary schools, adult education, cultural centers, women's centers (reversed).
Economic infrastructure	Distances to the two most popular microfinance offices and daily, three-day, five-day, weekly markets (reversed); processing, fish-smoking, shea-butter units. Higher = stronger infrastructure.
Security	Index of nearby terror attacks, oriented so higher = greater security.
Exogamy	Male, female, overall village exogamy. Male exogamy concerns outside grooms; female exogamy concerns women's out-marriage. These are distinct margins.

Baseline household/village reports underlie the first four indices; components are oriented before Anderson aggregation. The seven-predictor decomposition uses norms, SRH access, economic infrastructure, security, and three exogamy measures.

Tertiles and interactions. Tertiles use baseline village characteristics; tertile 1 is most constrained. Regressions include strata effects, moderator levels, and assignment interactions. Assignment coefficients give first-tertile effects; interaction coefficients are differences from them. Linear interaction tables standardize moderators. Context interactions characterize response differences, not causal returns to changing village conditions.

Measurement limits. Norms combine household and village information, not direct enforcement or price measures. Bride prices and match quality are unobserved; the framework's demand-side mechanism remains interpretive.

S2 Announcement margin and reach

The main paper reports the announcement index, participation splits, within-status arm comparisons, and household-head results. Here we retain the additional checks. The T3–T2 contrast combines the promise with induced participation, delivery, and salience differences; it does not isolate a psychological promise effect.

Table S5: The T2-vs-T3 contrast under familywise correction

Outcome	Wald p	Holm p (7-test family)	Holm p (all 26 tests)
Dis M. Early Any	0.011	0.077	0.286
Dis M. Early Nb	0.032	0.192	0.800
Adv M. Early Any	0.042	0.192	0.966
Adv M. Early Nb	0.080	0.240	1.000
Gender Att.	0.483	0.483	1.000
Prog. take up	0.032	0.192	0.800
Voc tr.	0.091	0.240	1.000

Notes: Wald p -values report (cluster-robust tests of $\beta_{T2} = \beta_{T3}$ from the four-arm ITT regressions). The seven-outcome family comprises the outcomes on which the announcement contrast rests—the four normative attitude outcomes, gender attitudes, program take-up, and vocational training—declared as a family before the joint correction was computed; because the declaration postdates the arm-level analyses, it is post-hoc, and the all-26 column reports the Holm correction over the complete set of T2-vs-T3 tests as the conservative benchmark. Holm step-down adjustments in both columns.

Multiplicity. The index-level T3–T2 margin is +0.265 SD (SE 0.109; RI $p = 0.029$). The index was constructed after observing the components. The smallest component-family adjusted p is about 0.08; no individual contrast survives a 5% familywise standard. T3-versus-control advocacy results and household-head contrasts provide separate corroboration with their own testing families in the main paper.

Reporting interpretation. INSD fielded the survey independently of implementers; the questionnaire did not discuss the grant and responses did not determine eligibility. The same arm pattern appears among household heads, whose outcomes have higher coverage. These checks constrain girl-specific classroom or reporting accounts, but cannot distinguish underlying normative change from community-wide affirmation of the announced program. No cash had been paid, excluding spillovers from grant receipt at this horizon.

Table S6 reports the underlying Wald tests of $H_0 : \beta_{T2} = \beta_{T3}$ outcome by outcome, from which the familywise corrections above are computed.

Table S6: Test of $H_0: \beta_{T2} = \beta_{T3}$ for Each Outcome

Outcome	F-statistic	p-value
<i>Panel A: Main Outcomes</i>		
Ever Married	0.07	0.791
Age at Mar.	1.94	0.165
Mar. bf18	2.61	0.108
Ever Pregnant	1.13	0.289
Curr. Pregnant	0.49	0.483
Preg. bf18	0.52	0.472
Emo. Violence	0.01	0.937
IGA	0.80	0.372
Mental Health	0.02	0.900
<i>Panel B: Intermediate Outcomes</i>		
Prog. take up	4.66	0.032**
Voc tr.	2.89	0.091*
Profes. tr.	0.77	0.383
Finance	0.38	0.536
Gender Att.	0.49	0.483
Aspirations	0.01	0.920
SRH Know.	0.22	0.637
SRH Behav.	0.49	0.487
MHM	1.42	0.235
Decision	0.37	0.543
Decision(C)	1.39	0.241
HH Opinion	0.17	0.682
Soc. Skills	0.30	0.586
Dis M. Early Any	6.69	0.011**
Dis M. Early Nb	4.66	0.032**
Adv M. Early Any	4.20	0.042**
Adv M. Early Nb	3.11	0.080*
Notes: Wald test from 4-arm ITT regression; H_0 : coefficient on T2 = coefficient on T3. Cluster-robust F-statistics. * p<0.10, ** p<0.05, *** p<0.01.		

Participation: baseline adjustment and composition

Table S7: Participation splits under ANCOVA (baseline attitude controlled)

Outcome	Non-participants	Participants	p (equal)	N
<i>Panel A: T3 vs. control (exposure split)</i>				
Dis M. Early Any	0.178** (0.071)	0.287*** (0.070)	0.027	2,596
Dis M. Early Nb	0.401** (0.189)	0.605*** (0.191)	0.106	2,596
Adv M. Early Any	-0.168*** (0.054)	-0.246*** (0.059)	0.032	2,596
Adv M. Early Nb	-0.353*** (0.112)	-0.484*** (0.129)	0.061	2,596
<i>Panel B: T3–T2 announcement margin (gap split)</i>				
Dis M. Early Any	0.128* (0.071)	0.180*** (0.062)	0.496	2,750
Dis M. Early Nb	0.274 (0.179)	0.428** (0.165)	0.402	2,750
Adv M. Early Any	-0.094* (0.052)	-0.099 (0.063)	0.924	2,750
Adv M. Early Nb	-0.176 (0.108)	-0.134 (0.122)	0.729	2,750

Notes: ANCOVA versions of the participation splits: each regression adds the baseline value of the outcome (zero-filled) and its missingness indicator to strata fixed effects and age dummies; village-clustered standard errors in parentheses. The baseline attitude variables are binary indicators (the baseline instrument recorded any-mention rather than counts), and enter as such. Panel A: coefficients on $T3 \times$ non-participant and $T3 \times$ participant indicators, T3-and-control sample. Panel B: same split of the T3–T2 contrast, pooled T2-and-T3 sample with a participant main effect. Participation is safe-space take-up (chosen, and itself shifted by the announcement), so the splits are descriptive; p (equal) tests equality of the two coefficients. Estimates are close to their non-ANCOVA counterparts in the main text, showing the arms’ baseline attitude differences do not account for the margins. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table S8: Composition-trimmed bounds on the within-status announcement margins

Outcome	Non-participants		Participants	
	untrimmed	trimmed bounds	untrimmed	trimmed bounds
Anti-early-marriage index	0.267	[0.035, 0.448]	0.354	[0.041, 0.768]
Dis M. Early Any	0.138	[0.059, 0.196]	0.202	[0.129, 0.398]
Dis M. Early Nb	0.295	[0.117, 0.584]	0.502	[−0.024, 1.085]
Adv M. Early Any	-0.105	[−0.150, −0.014]	-0.113	[−0.334, −0.030]
Adv M. Early Nb	-0.182	[−0.257, 0.058]	-0.153	[−0.528, −0.012]

Notes: Take-up responds to the announcement (33.3 percent of the T3 estimation sample participates against 24.3 percent in T2), so the participant and non-participant pools differ in composition across the two arms. Each cell bounds the within-status T3–T2 difference in unadjusted means by Lee-type trimming of the group whose pool contains the announcement-shifted margin: for the participant margin, the T3 participant distribution is trimmed from each tail by the induced share (27.1 percent of T3 participants); for the non-participant margin, the T2 non-participant distribution is trimmed by the corresponding share (11.9 percent). The non-participant margin remains one-sided on the index and on three of the four outcomes.

Trimming intervals are estimated identified sets, not confidence intervals. The main paper reports that the Imbens–Manski interval for the within-status index includes zero. Participation is chosen; these checks do not turn attendance into a randomized exposure.

The head-reported estimates in the paper are computed at the girl level, so a head linked to several roster girls enters once per girl. Table S9 places one observation per head alongside the published estimand.

Table S9: Household-head attitudes: girl-weighted and one observation per head

	Girl level (published)		One row per head	
	T3	SE	T3	SE
Head: any adv. mar. bf18	-0.168***	(0.059)	-0.159***	(0.059)
Head: any disadv. birth bf18	0.237***	(0.059)	0.224***	(0.061)
Head: any disadv. mar. bf18	0.225***	(0.062)	0.216***	(0.064)
Head: ideal marriage age	0.494***	(0.145)	0.416***	(0.149)

Notes: Coefficients on the announced-grant indicator, strata fixed effects, village-clustered standard errors. The girl-level columns reproduce the published estimand, in which a head linked to several roster girls enters once per girl; the head-level columns place one observation per head. Age dummies enter the girl-level specification only, age being a characteristic of the girl.

Attitude items

Table S10: Item-level decomposition of the early-marriage attitude counts

Item	Control mean	T2	T3	p_{RI} (T3)	Holm (T3)	T3 non-part.	T3 part.
Dis: STDs	0.084	-0.040	0.004	0.910	1.000	-0.001	0.026
Dis: child health	0.252	0.027	0.113**	0.062	0.372	0.108	0.139
Dis: domestic violence	0.239	0.039	0.095	0.122	0.509	0.078	0.127
Dis: loss of childhood	0.212	-0.003	0.117**	0.043	0.301	0.106	0.130
Dis: widowhood	0.064	0.015	0.053	0.147	0.509	0.049	0.069
Dis: no education	0.147	0.010	-0.021	0.619	1.000	-0.020	-0.013
Dis: mother's health	0.188	0.062	0.095*	0.102	0.509	0.080	0.126
Adv: having children	0.351	-0.102	-0.163***	0.010	0.090	-0.128	-0.205
Adv: finding a husband	0.317	-0.078	-0.166***	0.003	0.030	-0.153	-0.182
Adv: able to work	0.139	-0.082	-0.097**	0.021	0.168	-0.081	-0.120

Notes: Each row is one binary component of the early-marriage attitude counts (7 disadvantage items, 3 advantage items). T2/T3 columns: coefficients from a joint four-arm regression with strata fixed effects and age dummies, village-clustered SEs (stars on T3 analytic p -values). p_{RI} : randomization inference for the T3 coefficient, 1,000 within-strata permutations of the four-category assignment, $|t|$ criterion, plus-one convention; Holm: step-down correction across the ten items. The last two columns split the T3 effect by safe-space take-up (descriptive; participation is chosen). By linearity, the disadvantage-item T3 coefficients sum to the count ITT under the same specification. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The four normative outcomes are two pairs rather than four independent endpoints, so the index could be accused of counting the same response dimension twice. Table S11 reports the announcement margin under three alternative aggregations. The inverse-covariance procedure already places 87 percent of its weight on the two count components and 13 percent on the two any-mention indicators, and the margin is between 0.26 and 0.30 standard deviations under every weighting considered.

Table S11: The announcement margin under alternative index weightings

	T3–T2	SE	p	p_{RI}
Anderson (reported)	0.265**	(0.109)	0.017	0.030
Equal weight	0.284**	(0.111)	0.012	0.024
Binary items only	0.297***	(0.113)	0.010	0.018
Count items only	0.258**	(0.109)	0.020	0.033

Notes: Each row aggregates the same four normative outcomes, each oriented so that higher values are anti-early-marriage and standardized by the control group, and reports the coefficient on the announced-grant indicator in the pooled training-only and announced-grant sample, with the baseline outcomes and their missingness indicators, strata fixed effects and age dummies; village-clustered standard errors. The four components are two pairs rather than four independent endpoints: an any-mention indicator and a count from the disadvantage battery, and the same two from the advantage battery. Binary items only averages the two any-mention components; count items only averages the two counts. p_{RI} permutes the two arm labels among the villages actually assigned to them, within strata.

Grant eligibility and precision

Table S12: Grant-eligibility gradient of the T3 effect

Outcome	T3 effect by baseline age group			Eligibility gradient		Triple diff. vs T2	
	≤ 12	13	≥ 14	est.	p_{RI}	est.	p_{RI}
Index	0.634	0.747	0.476	-0.158	0.359	0.026	0.878
Dis M. Early Any	0.320	0.264	0.204	-0.116	0.224	0.097	0.301
Dis M. Early Nb	0.525	0.538	0.446	-0.078	0.722	0.219	0.294
Adv M. Early Any	-0.272	-0.336	-0.196	0.076	0.364	0.014	0.887
Adv M. Early Nb	-0.580	-0.762	-0.398	0.182	0.276	0.153	0.425

Notes: Grant eligibility required age 14 by program start; girls aged 12 or younger at baseline were still ineligible at program start, girls at 13 are ambiguous, and girls 14 or older were eligible. Columns 1–3: T3 coefficients by baseline-age group from a joint four-arm-by-group regression with age dummies and strata fixed effects, village-clustered standard errors (T3 cells: 75 / 87 / 1,357 respondents). Eligibility gradient: the eligible-minus-ineligible T3 contrast. Triple difference: that contrast net of the corresponding age gradient in the training-only arm (T2), which nets out age effects unrelated to eligibility. p_{RI} : randomization inference, 1,000 within-strata permutations of the four-category assignment. The design detects an eligibility contrast of roughly 0.20 on the binary outcomes at 80 percent power (design standard errors computed on pre-treatment baseline attitudes before estimation), so a flat gradient is informative against the strong form of the anticipation reading.

The 75 ineligible T3 respondents provide limited precision. Failure to reject an eligibility gradient does not demonstrate equal effects or exclude all anticipation mechanisms.

Attitudes and behavior: descriptive calibration

Table S13: Calibration of stated-attitude shifts to marriage and pregnancy margins

	Ever married by follow-up		Ever pregnant by follow-up	
	Control	All arms	Control	All arms
Names any disadvantage, baseline	0.105* (0.061)	0.037 (0.030)	0.009 (0.055)	0.030 (0.027)
Names any advantage, baseline	-0.060 (0.066)	-0.000 (0.029)	-0.071 (0.060)	0.016 (0.025)
Head: minimum marriage age for girls, baseline	-0.015*** (0.005)	-0.004* (0.003)	-0.008* (0.005)	-0.007** (0.003)
Implied change: T3 girl-attitude shifts	0.034	0.008	0.016	0.003
Implied change: T3 head ideal-age shift	-0.007	-0.002	-0.004	-0.003
Control mean	0.494	0.494	0.409	0.409
Observations	344	1,686	447	2,045

Descriptive, non-causal benchmark. Each column regresses the follow-up indicator on baseline attitude items among girls unmarried (never pregnant) at baseline, with strata fixed effects and baseline-age dummies; village-clustered standard errors in parentheses. Pooled columns add treatment-arm indicators. Implied changes multiply the T3 effects on the corresponding attitudes (disagreement +0.204, advocacy -0.206, head ideal marriage age +0.49 years) by the estimated slopes. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

This exercise maps attitude shifts through baseline predictive associations, not causal mediation. Implied girl-attitude marriage changes range from 0.8 to 3.4 percentage points; the head-age calibration is below one point. The main paper’s corresponding T3 marriage ITT among initially unmarried girls is 0.001 (SE 0.042). This illustrates limited precision; it does not establish that attitudes cause marriage outcomes or that the grant will eventually change behavior.

Table S14 places this design beside the two closest promise-based antecedents, separating what is promised, the condition attached to it, and the horizon at which outcomes are measured.

Table S14: Promise-based designs compared

	Apni Beti Apna Dhan, Haryana (Biswas and Das, 2026)	Buchmann et al. (2023), Bangladesh	This paper (Burkina Faso)
Transfer	Savings bond redeemable by the girl	In-kind incentive (cooking oil) to the household	\$400 grant to the participant
Conditionality	Payable at 18 if unmarried	Paid while the girl remains unmarried	Unconditional on marital status; tied to program participation
Horizon from announcement to payment	About a decade	Periodic over the program	About 18 months; unpaid at follow-up
Who is informed	Households at enrollment	Households and girls	Participants during training; community through local channels
Outcomes measured before any payment	No	No	Yes (all outcomes in this paper)
Stated attitudes or norms	No detectable change	Not the focus of the incentive arm	Large shift, concentrated on early-marriage items
Marriage timing	Child marriage reduced (long run)	Child marriage reduced	Null at follow-up

Notes: Entries summarize the designs as described in the papers cited in the introduction of the main text; the comparison is descriptive. The distinguishing feature of the present design is that every outcome was measured before any transfer had been paid, so the contrast between the announced-grant and training-only arms is a response to a credible promise rather than to receipt.

S3 Attrition, specification, and precision

Follow-up loss is substantial and partly marriage-related. Marginal Lee and covariate-conditional Semenova bounds address differential selection under monotonicity, not unrestricted recovery of the full cohort. Trimmed proportions are approximately 6.8% for T1 (control trimmed), 0.2% for T2, and 9.0% for T3 (treated trimmed), with item-specific variation. The target is the always-observed population. Treatment-induced displacement could violate monotonicity.

The assumption-free bounds below assign missing binary outcomes at logical extremes and span zero throughout. One-sided Lee-type identified sets are not confidence intervals excluding zero or full-cohort effects.

Table S15: Horowitz–Manski worst-case bounds for the binary main outcomes

Outcome	T1		T2		T3	
	LB	UB	LB	UB	LB	UB
Ever Married	-0.157	0.194	-0.232	0.157	-0.157	0.175
Mar. bf18	-0.325	0.411	-0.370	0.429	-0.397	0.316
Ever Pregnant	-0.402	0.377	-0.414	0.321	-0.336	0.338
Curr. Pregnant	-0.421	0.434	-0.417	0.413	-0.400	0.354
Preg. bf18	-0.499	0.493	-0.490	0.467	-0.489	0.411
Emo. Violence	-0.413	0.489	-0.396	0.477	-0.394	0.396

Notes: Worst-case (Horowitz–Manski) bounds on the arm–control difference in means, imputing every girl in the baseline roster with a missing follow-up outcome—whether from attrition or item nonresponse—at the logical extremes 0 and 1 in the adversarial direction. No selection assumption is imposed, so the bounds address the common follow-up loss of roughly 36 percent, not only the differential component that Lee-type trimming bounds handle. Unadjusted differences in means on the full baseline roster. The bounds bracket zero for every outcome and arm: absent any restriction on how attritors differ from respondents, the data alone cannot sign any effect, which is the sense in which the Lee and Semenova bounds reported alongside are informative only under their monotone-selection assumption.

Main outcomes: marginal selection bounds

Main outcomes: marginal selection bounds

Main outcomes: conditional selection bounds

Table S17: Generalized Lee Bounds (Semenova 2025) – Main Outcomes

Outcome	T1: Safe Spaces		T2: SS + Training		T3: SS + Training + Announced Grant	
	LB	UB	LB	UB	LB	UB
Ever Married	-0.1708	0.3595	-0.0327	-0.0318	-0.0046	0.0938
Age at Mar.	-1.1421	3.4775	-1.8054	2.1511	-0.1678	0.7423
Mar. bf18	-0.0288	0.1511	0.0197	0.1389	-0.0967	0.1311
Ever Pregnant	-0.0790	0.1158	-0.0845	0.0695	0.0329	0.2753
Curr. Pregnant	-0.0474	0.0246	-0.0341	0.0478	-0.0467	0.0419
Preg. bf18	-0.0412	0.0289	-0.0235	-0.0207	-0.1304	-0.0201
Emo. Violence	-0.0079	0.0953	0.0416	0.1229	-0.0055	0.1141
IGA	0.1630	0.3213	-0.0798	0.0633	-0.1179	0.3473
Mental Health	-0.1174	0.2219	-0.2834	-0.0382	-0.4193	0.0796

Notes: Sharp covariate-conditional Lee bounds (Semenova, 2025, *J. Econometrics* 251: 106055).

Instrument = treatment assignment; logistic post-LASSO first stage with outcome-specific LASSO controls and strata fixed effects.

Bounds condition on always-observed-with-outcome units (selection = not attrited AND outcome non-missing).

Where LB = UB: zero differential selection for this arm–outcome pair (point identified).

The bounding procedures need not deliver the same sign: they use different trimming/conditioning rules and outcome-specific samples. One-sided intervals for some behavioral outcomes do not recover effects among all baseline girls. Age-at-marriage bounds additionally concern an outcome observed only among girls who married.

Attitudes: differential-selection bounds

Table S18: Lee bounds for the four early-marriage attitude outcomes

Outcome	T1: Safe Spaces		T2: SS + Training		T3: SS + Training + Announced Grant	
	LB	UB	LB	UB	LB	UB
Dis M. Early Any	0.0034 [-0.0489, 0.0557]	0.0764 [0.0253, 0.1274]	0.0626 [0.0170, 0.1082]	0.0640 [0.0141, 0.1139]	0.2000 [0.1596, 0.2404]	0.2984 [0.2448, 0.3520]
Dis M. Early Nb	-0.0647 [-0.1982, 0.0688]	0.1399 [-0.0101, 0.2899]	0.0867 [-0.0416, 0.2149]	0.1075 [-0.1174, 0.3325]	0.2668 [0.1408, 0.3928]	0.6166 [0.4782, 0.7550]
Adv M. Early Any	-0.0949 [-0.1442, -0.0456]	-0.0219 [-0.0750, 0.0312]	-0.1054 [-0.1570, -0.0538]	-0.1040 [-0.1471, -0.0610]	-0.2881 [-0.3421, -0.2341]	-0.1897 [-0.2290, -0.1504]
Adv M. Early Nb	-0.1308 [-0.2274, -0.0343]	0.0338 [-0.1191, 0.1867]	-0.2426 [-0.3318, -0.1535]	-0.2301 [-0.3714, -0.0888]	-0.5846 [-0.6984, -0.4708]	-0.3608 [-0.4388, -0.2828]

Notes: Lee (2009) trimming bounds.

95% CIs in brackets below each bound, computed analytically.

The bounds address only *differential* selection between the arm and control; the trimmed share is the arm–control response differential as a share of the less-attrited group’s respondents (6.8% T1, 0.2% T2, 9.0% T3, varying marginally with item missingness); the $\approx 36\%$ follow-up loss common to all arms lies outside them, and the estimand is the always-observed subpopulation.

Table S19: Conditional Lee bounds for the four early-marriage attitude outcomes

Outcome	T1: Safe Spaces		T2: SS + Training		T3: SS + Training + Announced Grant	
	LB	UB	LB	UB	LB	UB
Dis M. Early Any	0.0183	0.1906	0.0628	0.0637	0.2000	0.2984
Dis M. Early Nb	-0.1416	0.5001	0.0746	0.1362	0.5150	1.2174
Adv M. Early Any	-0.1082	0.0605	-0.1075	0.0094	-0.2695	-0.0644
Adv M. Early Nb	-0.1979	0.2439	-0.2500	-0.2083	-0.4978	-0.0941

Notes: Sharp covariate-conditional Lee bounds (Semenova, 2025, *J. Econometrics* 251: 106055).

Instrument = treatment assignment; first stage uses a regularized logit with outcome-specific selected controls and strata fixed effects.

Where LB = UB: differential attrition rate is zero for this arm–outcome pair; bounds collapse to the ITT point estimate.

Interval endpoints are reported as (min, max); bounds condition on always-observed-with-outcome units.

Excerpts retain all four focal attitudes, all three arms, and reported endpoint confidence intervals. Other intermediate outcomes remain in the full technical supplement. Neither table resolves common follow-up loss.

Alternative control specifications

Table S20: Main ITT estimates under alternative control specifications

	Unadjusted		Prespecified		PDS-LASSO	
	Coef.	SE	Coef.	SE	Coef.	SE
<i>Panel A: T1 (Safe Spaces)</i>						
Ever Married	0.011	(0.025)	0.012	(0.017)	0.003	(0.017)
Age at Marriage	0.104	(0.154)	0.212*	(0.112)	0.253*	(0.146)
Married bf. 18	0.026	(0.038)	-0.004	(0.032)	-0.004	(0.034)
Ever Pregnant	-0.009	(0.050)	-0.010	(0.044)	-0.028	(0.049)
Curr. Pregnant	-0.017	(0.015)	-0.017	(0.015)	-0.020	(0.016)
Preg. bf18	-0.034	(0.033)	-0.011	(0.012)	-0.012	(0.012)
Emotional Violence	0.033	(0.032)	0.033	(0.031)	0.036	(0.033)
IGA	0.185	(0.161)	0.206	(0.160)	0.208	(0.167)
Mental Health	0.065	(0.156)	0.065	(0.157)	0.037	(0.160)
<i>Panel B: T2 (SS + Training)</i>						
Ever Married	-0.032	(0.028)	-0.005	(0.025)	-0.011	(0.026)
Age at Marriage	0.043	(0.172)	0.081	(0.109)	0.114	(0.153)
Married bf. 18	0.022	(0.038)	0.011	(0.032)	-0.006	(0.036)
Ever Pregnant	-0.071	(0.049)	-0.061	(0.047)	-0.080	(0.052)
Curr. Pregnant	-0.011	(0.016)	-0.010	(0.016)	-0.010	(0.017)
Preg. bf18	-0.030	(0.040)	0.000	(0.014)	-0.013	(0.014)
Emotional Violence	0.050*	(0.030)	0.048	(0.029)	0.057*	(0.033)
IGA	-0.029	(0.142)	-0.012	(0.139)	-0.098	(0.135)
Mental Health	-0.113	(0.168)	-0.113	(0.168)	-0.140	(0.180)
<i>Panel C: T3 (SS + Training + Announced Grant)</i>						
Ever Married	0.015	(0.024)	0.003	(0.020)	0.006	(0.020)
Age at Marriage	0.323**	(0.154)	0.239**	(0.113)	0.263**	(0.133)
Married bf. 18	-0.048	(0.033)	-0.039	(0.029)	-0.045	(0.031)
Ever Pregnant	-0.001	(0.045)	-0.007	(0.041)	-0.018	(0.046)
Curr. Pregnant	0.005	(0.017)	0.003	(0.017)	0.006	(0.017)
Preg. bf18	-0.045	(0.031)	-0.010	(0.011)	-0.013	(0.012)
Emotional Violence	0.038	(0.029)	0.048*	(0.029)	0.051*	(0.030)
IGA	0.075	(0.150)	0.084	(0.148)	0.021	(0.151)
Mental Health	-0.117	(0.147)	-0.116	(0.147)	-0.076	(0.150)

Notes: Each cell reports the coefficient on the arm indicator, village-clustered SEs in parentheses. Unadjusted: strata fixed effects only. Prespecified: strata fixed effects, age dummies, and the baseline outcome with its missingness indicator where available. PDS-LASSO: a reimplementation of the published post-double-selection specification (selection over the 65 candidate baseline controls, partialling strata fixed effects and the baseline outcome) within this sensitivity script; because the lasso selection step is re-run here, coefficients can differ modestly from the published estimates (e.g., T3 IGA 0.021 here versus 0.114 in the main-paper ITT table), and the main-paper ITT table remains the authoritative set. Sample: non-attributed analysis sample; Preg. bf18 is unconditional (all non-attributed girls; D011b). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The analysis roster is broader than the program’s own eligibility rule, which admitted out-of-school girls aged 10 to 19 and married girls and young women under 24. Table S21 re-estimates the main effects on the eligible subsample. The survey file records whether a girl had ever attended school but not whether she was enrolled at baseline, so the eligible column is built from age and baseline marital status alone; a stricter cut that also drops girls who had ever attended school bounds the schooling condition from the other side.

Table S21: Main ITT estimates on the program-eligible subsample

	Full roster		Eligible		Eligible, never schooled	
	T3	N	T3	N	T3	N
Age at Marriage	0.239**	3686	0.250**	3469	0.305**	2866
Curr. Pregnant	0.003	4601	0.004	4267	0.011	3543
Emotional Violence	0.048*	3951	0.042	3611	0.030	2992
Ever Married	0.003	4935	0.003	4575	0.003	3814
Ever Pregnant	-0.007	4936	-0.012	4575	-0.009	3814
IGA	0.084	4935	0.088	4574	0.069	3814
Married bf. 18	-0.039	3597	-0.032	3359	-0.043	2760
Mental Health	-0.116	4936	-0.085	4575	-0.037	3814
Preg. bf18	-0.010	4087	-0.001	3730	-0.003	3075

Notes: Coefficients on the announced-grant indicator from the prespecified specification (strata fixed effects, age dummies, and the baseline outcome with its missingness indicator where available); village-clustered standard errors. The program’s own rule admitted out-of-school girls aged 10 to 19 and married girls and young women under 24. The survey file records whether a girl had ever attended school but not whether she was enrolled at baseline, so the eligible column is built from age and baseline marital status alone; the third pair of columns additionally drops girls who had ever attended school, bounding the schooling condition from the other side.

Most of the roster had already experienced the events the marriage and fertility outcomes measure, so the average effects are taken largely over girls whose outcome could not change. Table S22 restricts each margin to the girls still at risk at baseline and repeats the full-roster estimates underneath for comparison. The baseline outcome is omitted from the at-risk rows because it is degenerate there, every girl in the restricted sample having the same baseline value.

Table S22: Marriage and pregnancy among the girls still at risk at baseline

	T1 Safe spaces	T2 SS + Training	T3 SS+Tr.+ Ann. Grant	Control mean	N
Ever married, unmarried at baseline	-0.033 (0.037) [0.417]	-0.020 (0.048) [0.701]	0.001 (0.042) [0.982]	0.494	1686
Ever married, full roster	0.012 (0.018) [0.494]	-0.026 (0.023) [0.269]	0.008 (0.018) [0.646]	0.799	4935
Ever pregnant, never pregnant at baseline	0.027 (0.035) [0.459]	-0.004 (0.041) [0.914]	0.001 (0.041) [0.970]	0.409	2045
Ever pregnant, full roster	-0.007 (0.043) [0.884]	-0.064 (0.047) [0.207]	-0.006 (0.041) [0.872]	0.631	4936

Notes: Coefficients on the arm indicators, with village-clustered standard errors in parentheses and randomization-inference p -values in brackets from 1,000 within-strata permutations of the four-category village assignment. Strata fixed effects and age dummies throughout. The at-risk rows restrict each margin to the girls who had not yet experienced the event at baseline; the baseline outcome is omitted from those rows because it is degenerate within the restricted sample. The full-roster rows repeat the same specification on all non-attrited girls and are shown for comparison. Control means are taken within the sample of the row. Stars mark randomization-inference significance: $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Compliance-adjusted magnitudes

The tables below rescale the intent-to-treat estimates by the arm-specific first stage. They are Wald-scaled ITTs, not complier average effects: assignment changes a bundle, while the take-up indicator records safe-space participation alone, so the exclusion restriction required for a local-average-treatment-effect reading is not defensible here.

Table S23: Compliance-Adjusted Magnitudes (Wald-Scaled ITTs), Main Outcomes

Outcome	T1: Safe Spaces			T2: SS + Training			T3: SS + Training + Announced Grant		
	2SLS	SE	1st-F	2SLS	SE	1st-F	2SLS	SE	1st-F
Ever Married	0.046	(0.066)	55.7	-0.031	(0.124)	42.1	0.009	(0.071)	53.8
Age at Mar.	0.650*	(0.362)	56.7	0.376	(0.420)	42.5	0.881**	(0.402)	53.4
Mar. bf18	-0.008	(0.101)	58.9	0.080	(0.146)	41.2	-0.146	(0.104)	53.5
Ever Pregnant	-0.047	(0.150)	55.4	-0.286	(0.226)	40.8	-0.017	(0.132)	53.6
Curr. Pregnant	-0.068	(0.057)	54.6	-0.031	(0.082)	40.9	-0.002	(0.055)	53.6
Preg. bf18	-0.027	(0.039)	55.4	-0.016	(0.058)	40.8	-0.052	(0.040)	53.6
Emo. Violence [†]	0.130	(0.112)	52.8	0.228*	(0.137)	43.2	0.218**	(0.098)	51.3
IGA	0.780	(0.612)	53.6	-0.156	(0.657)	40.9	0.438	(0.505)	53.5
Mental Health [†]	0.319	(0.546)	54.7	-0.769	(0.808)	41.2	-0.348	(0.474)	53.6

Notes: 2SLS estimates. Instrument = treatment assignment (arm-specific).

Sample restricted per arm: Tk union control (non-attrited). Same LASSO-selected controls as ITT.

1st-F = first-stage F-statistic. * p<0.10, ** p<0.05, *** p<0.01.

All entries are compliance-adjusted magnitudes (numerically, the ITT scaled by the arm-specific first stage), not causal complier effects: assignment changes a bundle of components and exposures beyond the measured take-up indicator, so the exclusion restriction underlying a LATE interpretation is not maintained for any outcome. [†] For emotional violence and mental health, non-participants in treated villages are themselves partially exposed, so the rescaled figure is best read as an upper bound on the per-participant effect (see the main text).

Table S24: Heterogeneous Compliance-Adjusted Magnitudes (T3) by Social-Norms Permissiveness Tertile

Outcome	Tertile 1 (Restrictive)		Tertile 2		Tertile 3 (Permissive)	
	LATE	p	LATE	p	LATE	p
Ever Married	0.182	0.753	-0.088	0.229	0.234	0.382
Age at Mar.	-0.476	0.861	0.958**	0.024	1.760**	0.034
Mar. bf18	0.361	0.614	-0.201	0.159	0.086	0.697
Ever Pregnant	-1.299	0.245	-0.115	0.290	0.528	0.275
Curr. Pregnant	-0.204	0.503	0.072	0.272	-0.071	0.569
Preg. bf18	-0.180	0.600	-0.076*	0.051	0.063	0.613
Emo. Violence	0.457	0.378	0.215	0.101	-0.020	0.873
IGA	2.597	0.290	1.249*	0.061	-1.574	0.102
Mental Health	-7.423*	0.092	0.330	0.541	-0.842	0.469

Notes: 2SLS (T3 only). Instrument = T3. Tertiles of the baseline village social-norms permissiveness index (Unrestricted). Tertile 1 = most restrictive/conservative villages. * p<0.10, ** p<0.05, *** p<0.01.

Table S25: Heterogeneous Compliance-Adjusted Magnitudes (T3) by SRH-Access Tertile

Outcome	Tertile 1 (Least Access)		Tertile 2		Tertile 3 (Most Access)	
	LATE	p	LATE	p	LATE	p
Ever Married	-0.039	0.845	0.093	0.536	0.003	0.973
Age at Mar.	-0.613	0.453	1.835***	0.002	0.088	0.883
Mar. bf18	0.164	0.485	-0.128	0.398	-0.204	0.303
Ever Pregnant	-0.106	0.733	-0.255	0.280	0.260*	0.074
Curr. Pregnant	0.179*	0.058	-0.178*	0.080	0.171**	0.024
Preg. bf18	-0.007	0.935	-0.069	0.361	-0.099	0.197
Emo. Violence	0.204	0.174	0.025	0.846	0.261	0.177
IGA	-0.378	0.705	-0.886	0.180	0.393	0.516
Mental Health	-2.824*	0.093	-0.761	0.275	1.537**	0.045

Notes: 2SLS (T3 only). Instrument = T3. Tertiles of the baseline village SRH-access index.
Tertile 1 = least accessible villages. * p<0.10, ** p<0.05, *** p<0.01.

The T2 baseline marriage imbalance

Table S26 reports the two checks described in the paper: adding the village-level baseline marriage rate as a control, and restricting both T2 and control to girls never married at baseline so that the comparison rests on common support.

Table S26: T2 Coefficient Robustness to Baseline Marriage Imbalance

Outcome	Baseline ITT		+ Village Mar. Rate		Never-Mar. at BL	
	T2	SE	T2	SE	T2	SE
Ever Married	0.019	(0.036)	-0.012	(0.027)	0.041	(0.077)
Age at Mar.	0.043	(0.172)	-0.017	(0.172)	0.243	(0.199)
Mar. bf18	0.022	(0.038)	0.035	(0.039)	-0.028	(0.075)
Ever Pregnant	-0.057	(0.050)	-0.053	(0.050)	-0.004	(0.067)
Curr. Pregnant	-0.011	(0.016)	-0.007	(0.016)	-0.026	(0.019)
Preg. bf18	0.006	(0.016)	-0.000	(0.016)	-0.011	(0.042)

Notes: ANCOVA with baseline outcome control and strata FE.

Col. 3–4: village-level baseline ever-married rate added as additional control.

Col. 5–6: T2 and control samples restricted to women never-married at baseline.

* p<0.10, ** p<0.05, *** p<0.01.

Precision and the role of sample loss

Table S27: Ex-Post Minimum Detectable Effects for the Main Outcomes

	T1: Safe	T2: SS +	T3:	Control	MDE/
Outcome	Spaces	Training	SS+Tr.+ Ann. Grant	Mean	Mean (T3, %)
Ever Married	0.048	0.070	0.056	0.799	7.0
Age at Mar.	0.300	0.288	0.308	17.656	–
Mar. bf18	0.084	0.087	0.078	0.400	19.6
Ever Pregnant	0.118	0.126	0.109	0.631	17.3
Curr. Pregnant	0.042	0.045	0.048	0.086	55.3
Preg. bf18	0.031	0.036	0.031	0.214	14.4
Emo. Violence	0.090	0.081	0.084	0.085	98.8
IGA	0.456	0.386	0.414	-0.007	–
Mental Health	0.440	0.470	0.412	0.003	–

Notes: Ex-post minimum detectable effects at 80 percent power for 5 percent two-sided tests, computed as $2.8 \times$ the village-clustered standard error of the PDS-LASSO ITT estimates (the main-paper ITT table). The final column expresses the T3 MDE as a share of the control mean for binary outcomes.

Precision counterfactual. The post-mortem exercise holds estimates, outcome dispersion, and within-village correlation fixed and rescales standard errors by

$$\frac{SE_f}{SE_r} = \sqrt{\frac{J_r m_r}{J_f m_f} \frac{1 + (m_f - 1)\rho}{1 + (m_r - 1)\rho}},$$

where J denotes villages, m mean cluster size, ρ intra-cluster correlation, and r, f the realized and full-baseline designs. With 94% of villages retained, reported factors are 0.89–0.96: restoring all girls would reduce standard errors by 4–11% under these assumptions. Baseline-dispersion sensitivity gives factors of 0.77–1.06. These are precision calculations, not selection corrections; rescaled p -values do not replace re-running RI and Holm. Detailed counterfactual tables remain in the full technical record.

Nulls. MDEs describe detectable effects at stated power. They neither prove zero effects nor repair attrition. Incomplete delivery, low take-up, common outreach, and effects below detectable magnitudes remain distinct explanations for incremental behavioral nulls.

The coefficient-by-coefficient counterfactual follows. Point estimates are held at their realized values throughout, so the exercise isolates precision and says nothing about whether attrition moved the estimates themselves.

Table S28: Post-Mortem Precision Counterfactual: Main Outcomes (ITT)

Outcome	T1: Safe Spaces	T2: SS + Training	T3: SS+Tr.+ Ann. Grant	Control Mean	N
Ever Married	0.010 (0.017) [0.016] [-0.021, 0.041]	-0.006 (0.025) [0.023] [-0.051, 0.039]	0.001 (0.020) [0.018] [-0.035, 0.037]	0.799	4935
<i>implied p (A)</i>	[0.523]	[0.794]	[0.957]		
<i>implied p (B)</i>	[0.481]	[0.774]	[0.952]		
Age at Mar.	0.185 (0.107) [0.099] [-0.011, 0.381]	0.061 (0.103) [0.095] [-0.127, 0.249]	0.224 (0.110) [0.102] [0.023, 0.425]	17.656	3686
<i>implied p (A)</i>	[0.064]	[0.524]	[0.029]		
<i>implied p (B)</i>	[0.039]	[0.477]	[0.015]		
Mar. bf18	-0.004 (0.030) [0.028] [-0.059, 0.051]	0.016 (0.031) [0.029] [-0.040, 0.072]	-0.033 (0.028) [0.026] [-0.084, 0.018]	0.400	3597
<i>implied p (A)</i>	[0.885]	[0.576]	[0.202]		
<i>implied p (B)</i>	[0.874]	[0.540]	[0.162]		
Ever Pregnant	-0.009 (0.042) [0.040] [-0.088, 0.070]	-0.056 (0.045) [0.043] [-0.140, 0.028]	-0.006 (0.039) [0.037] [-0.079, 0.067]	0.631	4936
<i>implied p (A)</i>	[0.822]	[0.192]	[0.871]		
<i>implied p (B)</i>	[0.828]	[0.208]	[0.876]		
Curr. Pregnant	-0.018 (0.015) [0.013] [-0.044, 0.008]	-0.011 (0.016) [0.014] [-0.039, 0.017]	0.001 (0.017) [0.015] [-0.029, 0.031]	0.086	4601
<i>implied p (A)</i>	[0.180]	[0.442]	[0.947]		
<i>implied p (B)</i>	[0.159]	[0.419]	[0.945]		
Preg. bf18	-0.009 (0.011) [0.010] [-0.029, 0.011]	-0.002 (0.013) [0.012] [-0.026, 0.022]	-0.012 (0.011) [0.010] [-0.032, 0.008]	0.214	4087
<i>implied p (A)</i>	[0.383]	[0.870]	[0.245]		
<i>implied p (B)</i>	[0.257]	[0.831]	[0.131]		
Emo. Violence	0.032 (0.032) [0.030] [-0.028, 0.092]	0.049 (0.029) [0.027] [-0.005, 0.103]	0.051 (0.030) [0.028] [-0.005, 0.107]	0.085	3951
<i>implied p (A)</i>	[0.292]	[0.076]	[0.074]		
<i>implied p (B)</i>	[0.306]	[0.085]	[0.083]		
IGA	0.206 (0.163) [0.156] [-0.103, 0.515]	-0.010 (0.138) [0.132] [-0.271, 0.251]	0.114 (0.148) [0.142] [-0.166, 0.394]	-0.007	4935
<i>implied p (A)</i>	[0.189]	[0.940]	[0.423]		
<i>implied p (B)</i>	[0.180]	[0.939]	[0.414]		
Mental Health	0.073 (0.157) [0.151] [-0.225, 0.371]	-0.111 (0.168) [0.161] [-0.430, 0.208]	-0.090 (0.147) [0.141] [-0.369, 0.189]	0.003	4936
<i>implied p (A)</i>	[0.629]	[0.493]	[0.525]		
<i>implied p (B)</i>	n/a	n/a	n/a		

Notes: Point estimates at their realized values. Round parentheses: realized cluster-robust standard error (4,936 girls, 162 villages). Square brackets: counterfactual standard error had the full baseline sample been retained (7,719 girls, 173 villages), Method A: $SE_{full} = SE_{real} \sqrt{\frac{J_r m_r}{J_f m_f} \cdot \frac{1+(m_f-1)\rho}{1+(m_r-1)\rho}}$ at the realized endline ICC. Braces: 95% CI from the Method-A standard error with $t(J_f - 4)$ critical values. Implied p rows are two-sided $t(J_f - 4)$ p -values under Methods A and B (B additionally corrects the SD and ICC to their full-baseline-sample values; n/a where no baseline measure exists). The published inference is randomization inference, which cannot be rescaled analytically; read this table through SEs, CIs, and MDEs. No significance stars are used.

Table S29: Post-Mortem Precision Counterfactual: Intermediate Outcomes (ITT), Panel A

Outcome	T1: Safe Spaces	T2: SS + Training	T3: SS+Tr.+ Ann. Grant	Control Mean	N
Prog. take up	0.253 (0.040) [0.038] [0.177, 0.329]	0.198 (0.036) [0.034] [0.130, 0.266]	0.293 (0.040) [0.038] [0.217, 0.369]	0.039	4936
<i>implied p (A)</i>	[0.000]	[0.000]	[0.000]		
<i>implied p (B)</i>	n/a	n/a	n/a		
Voc tr.	0.020 (0.021) [0.020] [-0.019, 0.059]	0.067 (0.025) [0.024] [0.020, 0.114]	0.114 (0.028) [0.027] [0.061, 0.167]	0.038	4936
<i>implied p (A)</i>	[0.318]	[0.005]	[0.000]		
<i>implied p (B)</i>	n/a	n/a	n/a		
Profes. tr.	0.028 (0.026) [0.025] [-0.020, 0.076]	0.063 (0.021) [0.020] [0.024, 0.102]	0.075 (0.021) [0.020] [0.036, 0.114]	0.040	4908
<i>implied p (A)</i>	[0.256]	[0.002]	[0.000]		
<i>implied p (B)</i>	n/a	n/a	n/a		
Finance	0.037 (0.110) [0.105] [-0.170, 0.244]	0.049 (0.134) [0.128] [-0.203, 0.301]	0.119 (0.103) [0.098] [-0.075, 0.313]	0.003	4936
<i>implied p (A)</i>	[0.724]	[0.701]	[0.227]		
<i>implied p (B)</i>	[0.712]	[0.689]	[0.206]		
Gender Att.	0.046 (0.119) [0.114] [-0.179, 0.271]	0.323 (0.121) [0.116] [0.094, 0.552]	0.428 (0.118) [0.113] [0.205, 0.651]	-0.005	4935
<i>implied p (A)</i>	[0.687]	[0.006]	[0.000]		
<i>implied p (B)</i>	[0.686]	[0.006]	[0.000]		
Aspirations	-0.200 (0.117) [0.112] [-0.421, 0.021]	0.109 (0.129) [0.123] [-0.134, 0.352]	0.134 (0.125) [0.119] [-0.102, 0.370]	0.000	4936
<i>implied p (A)</i>	[0.075]	[0.378]	[0.264]		
<i>implied p (B)</i>	[0.092]	[0.404]	[0.290]		
SRH Know.	0.120 (0.159) [0.153] [-0.182, 0.422]	0.103 (0.149) [0.143] [-0.180, 0.386]	0.175 (0.140) [0.134] [-0.090, 0.440]	0.000	4136
<i>implied p (A)</i>	[0.433]	[0.473]	[0.195]		
<i>implied p (B)</i>	[0.409]	[0.449]	[0.172]		
SRH Behav.	0.068 (0.130) [0.124] [-0.177, 0.313]	0.175 (0.108) [0.103] [-0.028, 0.378]	0.157 (0.107) [0.102] [-0.044, 0.358]	-0.006	4136
<i>implied p (A)</i>	[0.584]	[0.091]	[0.126]		
<i>implied p (B)</i>	[0.558]	[0.071]	[0.102]		
MHM	0.033 (0.107) [0.103] [-0.170, 0.236]	-0.062 (0.102) [0.098] [-0.255, 0.131]	0.043 (0.100) [0.096] [-0.146, 0.232]	0.539	4181
<i>implied p (A)</i>	[0.748]	[0.527]	[0.655]		
<i>implied p (B)</i>	[0.716]	[0.474]	[0.612]		

Notes: Point estimates at their realized values. Round parentheses: realized cluster-robust standard error (4,936 girls, 162 villages). Square brackets: counterfactual standard error had the full baseline sample been retained (7,719 girls, 173 villages), Method A: $SE_{full} = SE_{real} \sqrt{\frac{J_r m_r}{J_f m_f} \cdot \frac{1+(m_f-1)\rho}{1+(m_r-1)\rho}}$ at the realized endline ICC. Braces: 95% CI from the Method-A standard error with $t(J_f - 4)$ critical values. Implied p rows are two-sided $t(J_f - 4)$ p -values under Methods A and B (B additionally corrects the SD and ICC to their full-baseline-sample values; n/a where no baseline measure exists). The published inference is randomization inference, which cannot be rescaled analytically; read this table through SEs, CIs, and MDEs. No significance stars are used.

Table S30: Post-Mortem Precision Counterfactual: Intermediate Outcomes (ITT), Panel B

Outcome	T1: Safe Spaces	T2: SS + Training	T3: SS+Tr.+ Ann. Grant	Control Mean	N
Decision	0.062 (0.152) [0.146] [-0.227, 0.351]	0.032 (0.138) [0.133] [-0.230, 0.294]	-0.073 (0.142) [0.137] [-0.343, 0.197]	0.007	4936
<i>implied p (A)</i>	[0.672]	[0.810]	[0.594]		
<i>implied p (B)</i>	[0.676]	[0.812]	[0.599]		
Decision(C)	0.163 (0.203) [0.194] [-0.219, 0.545]	0.181 (0.143) [0.136] [-0.088, 0.450]	0.007 (0.151) [0.144] [-0.277, 0.291]	0.002	2942
<i>implied p (A)</i>	[0.401]	[0.186]	[0.961]		
<i>implied p (B)</i>	[0.389]	[0.175]	[0.960]		
HH Opinion	-0.070 (0.153) [0.147] [-0.360, 0.220]	-0.102 (0.168) [0.161] [-0.421, 0.217]	-0.056 (0.165) [0.159] [-0.369, 0.257]	0.001	4917
<i>implied p (A)</i>	[0.635]	[0.528]	[0.724]		
<i>implied p (B)</i>	[0.631]	[0.524]	[0.722]		
Soc. Skills	-0.066 (0.164) [0.158] [-0.378, 0.246]	0.034 (0.176) [0.170] [-0.301, 0.369]	-0.034 (0.173) [0.167] [-0.363, 0.295]	0.011	4936
<i>implied p (A)</i>	[0.677]	[0.841]	[0.839]		
<i>implied p (B)</i>	n/a	n/a	n/a		
Dis M. Early Any	0.032 (0.080) [0.077] [-0.120, 0.184]	0.055 (0.077) [0.074] [-0.091, 0.201]	0.204 (0.070) [0.067] [0.071, 0.337]	0.521	4935
<i>implied p (A)</i>	[0.677]	[0.458]	[0.003]		
<i>implied p (B)</i>	[0.672]	[0.450]	[0.002]		
Dis M. Early Nb	0.053 (0.212) [0.203] [-0.348, 0.454]	0.112 (0.200) [0.192] [-0.267, 0.491]	0.436 (0.184) [0.176] [0.088, 0.784]	1.184	4935
<i>implied p (A)</i>	[0.795]	[0.560]	[0.014]		
<i>implied p (B)</i>	[0.792]	[0.555]	[0.013]		
Adv M. Early Any	-0.057 (0.080) [0.077] [-0.208, 0.094]	-0.100 (0.067) [0.064] [-0.227, 0.027]	-0.206 (0.058) [0.056] [-0.316,-0.096]	0.439	4935
<i>implied p (A)</i>	[0.458]	[0.121]	[0.000]		
<i>implied p (B)</i>	[0.450]	[0.115]	[0.000]		
Adv M. Early Nb	-0.127 (0.170) [0.163] [-0.449, 0.195]	-0.237 (0.146) [0.140] [-0.514, 0.040]	-0.414 (0.128) [0.123] [-0.657,-0.171]	0.807	4935
<i>implied p (A)</i>	[0.438]	[0.093]	[0.001]		
<i>implied p (B)</i>	[0.413]	[0.077]	[0.000]		

Notes: Point estimates at their realized values. Round parentheses: realized cluster-robust standard error (4,936 girls, 162 villages). Square brackets: counterfactual standard error had the full baseline sample been retained (7,719 girls, 173 villages), Method A: $SE_{full} = SE_{real} \sqrt{\frac{J_f m_f}{J_f m_r} \cdot \frac{1+(m_f-1)\rho}{1+(m_r-1)\rho}}$ at the realized endline ICC. Braces: 95% CI from the Method-A standard error with $t(J_f - 4)$ critical values. Implied p rows are two-sided $t(J_f - 4)$ p -values under Methods A and B (B additionally corrects the SD and ICC to their full-baseline-sample values; n/a where no baseline measure exists). The published inference is randomization inference, which cannot be rescaled analytically; read this table through SEs, CIs, and MDEs. No significance stars are used.

Table S31: Design-Based Minimum Detectable Effects: Realized vs. Full-Sample Counterfactual (T3 vs. Control)

Outcome	Table	MDE (native units)		MDE (SD units)		ρ (real.)	SE ratio (A)
		Realized	Full	Realized	Full		
Ever Married	Main	0.077	0.071	0.192	0.178	0.067	0.918
Age at Mar.	Main	0.512	0.479	0.238	0.223	0.111	0.927
Mar. bf18	Main	0.111	0.103	0.226	0.210	0.095	0.920
Ever Pregnant	Main	0.142	0.136	0.290	0.278	0.197	0.949
Curr. Pregnant	Main	0.044	0.040	0.164	0.147	0.037	0.891
Preg. bf18	Main	0.099	0.093	0.251	0.237	0.133	0.936
Emo. Violence	Main	0.094	0.089	0.287	0.274	0.185	0.946
IGA	Main	0.415	0.401	0.379	0.366	0.359	0.959
Mental Health	Main	0.429	0.416	0.410	0.398	0.427	0.961
Prog. take up	Intermediate	0.148	0.143	0.348	0.336	0.297	0.957
Voc tr.	Intermediate	0.091	0.087	0.302	0.290	0.216	0.951
Profes. tr.	Intermediate	0.076	0.073	0.265	0.253	0.158	0.945
Finance	Intermediate	0.344	0.331	0.309	0.297	0.227	0.952
Gender Att.	Intermediate	0.345	0.334	0.354	0.342	0.310	0.957
Aspirations	Intermediate	0.308	0.297	0.340	0.328	0.282	0.956
SRH Know.	Intermediate	0.382	0.371	0.409	0.397	0.422	0.961
SRH Behav.	Intermediate	0.290	0.279	0.327	0.314	0.253	0.953
MHM	Intermediate	0.303	0.293	0.404	0.391	0.410	0.960
Decision	Intermediate	0.395	0.384	0.420	0.407	0.449	0.962
Decision(C)	Intermediate	0.438	0.425	0.423	0.410	0.445	0.953
HH Opinion	Intermediate	0.409	0.397	0.405	0.393	0.416	0.961
Soc. Skills	Intermediate	0.425	0.413	0.459	0.446	0.543	0.964
Dis M. Early Any	Intermediate	0.186	0.180	0.382	0.370	0.367	0.959
Dis M. Early Nb	Intermediate	0.491	0.475	0.379	0.367	0.361	0.959
Adv M. Early Any	Intermediate	0.173	0.167	0.366	0.354	0.333	0.958
Adv M. Early Nb	Intermediate	0.356	0.345	0.393	0.380	0.389	0.960

Notes: MDEs at $\alpha = .05$ (two-sided), power 0.80: $\text{MDE} = 2.8016 \sigma \sqrt{[1 + (m - 1)\rho] (1/(J_{T3}m) + 1/(J_Cm))}$, unadjusted for covariates (conservative). Both scenarios hold the outcome SD and ICC at their realized endline values; only the design changes. Realized: $J_{T3} = 46$, $J_C = 38$ villages at the realized mean cluster size. Full-sample counterfactual: $J_{T3} = 48$, $J_C = 40$ villages at the counterfactual cluster size. SD units divide by the realized σ . SE ratio (A) is the Method-A counterfactual-to-realized standard-error ratio (applies to all arms).

Table S32: Post-Mortem Parameters by Outcome

Outcome	Table	Realized (endline, non-attrited)					Full baseline sample					b
		N	J	m	σ	ρ	N	J	m	σ_0	ρ_0	
Ever Married	Main	4935	162	30.5	0.402	0.067	7717	173	44.6	0.489	0.195	0.907
Age at Mar.	Main	3686	162	22.8	2.153	0.111	5764	173	33.3	8.711	0.165	0.896
Mar. bf18	Main	3597	162	22.2	0.490	0.095	5625	173	32.5	0.445	0.104	0.912
Ever Pregnant	Main	4936	162	30.5	0.488	0.197	7719	173	44.6	0.499	0.087	1.037
Curr. Pregnant	Main	4601	162	28.4	0.270	0.037	7195	173	41.6	0.206	0.036	0.952
Preg. bf18	Main	4087	162	25.2	0.393	0.133	6391	173	36.9	0.250	0.072	0.769
Emo. Violence	Main	3951	162	24.4	0.326	0.185	5913	173	34.2	0.340	0.256	1.030
IGA	Main	4935	162	30.5	1.096	0.359	7717	173	44.6	0.890	0.349	0.980
Mental Health	Main	4936	162	30.5	1.045	0.427	7719	173	44.6	n/a	n/a	n/a
Prog. take up	Intermediate	4936	162	30.5	0.427	0.297	7719	173	44.6	n/a	n/a	n/a
Voc tr.	Intermediate	4936	162	30.5	0.301	0.216	7719	173	44.6	n/a	n/a	n/a
Profes. tr.	Intermediate	4908	162	30.3	0.289	0.158	7675	173	44.4	n/a	n/a	n/a
Finance	Intermediate	4936	162	30.5	1.116	0.227	7719	173	44.6	0.978	0.190	0.956
Gender Att.	Intermediate	4935	162	30.5	0.975	0.310	7717	173	44.6	0.928	0.324	0.998
Aspirations	Intermediate	4936	162	30.5	0.908	0.282	7719	173	44.6	1.000	0.153	1.057
SRH Know.	Intermediate	4136	162	25.5	0.933	0.422	6190	173	35.8	1.003	0.381	0.949
SRH Behav.	Intermediate	4136	162	25.5	0.887	0.253	6190	173	35.8	0.960	0.286	0.935
MHM	Intermediate	4181	162	25.8	0.749	0.410	6538	173	37.8	0.967	0.307	0.882
Decision	Intermediate	4936	162	30.5	0.942	0.449	7719	173	44.6	1.064	0.442	1.014
Decision(C)	Intermediate	2942	156	18.9	1.037	0.445	4217	169	25.0	0.807	0.407	0.975
HH Opinion	Intermediate	4917	162	30.4	1.010	0.416	7689	173	44.4	1.102	0.602	0.990
Soc. Skills	Intermediate	4936	162	30.5	0.927	0.543	7719	173	44.6	n/a	n/a	n/a
Dis M. Early Any	Intermediate	4935	162	30.5	0.487	0.367	7717	173	44.6	0.487	0.348	0.983
Dis M. Early Nb	Intermediate	4935	162	30.5	1.294	0.361	7717	173	44.6	0.484	0.347	0.986
Adv M. Early Any	Intermediate	4935	162	30.5	0.472	0.333	7717	173	44.6	0.487	0.336	0.983
Adv M. Early Nb	Intermediate	4935	162	30.5	0.906	0.389	7717	173	44.6	0.385	0.287	0.949

Notes: N = observations, J = villages, $m = N/J$. Realized: endline outcome on the non-attrited sample (N = published regression N). Full baseline sample: σ_0, ρ_0 measured on the baseline analogue of the outcome over all baseline respondents; N scales the realized N by the outcome's baseline coverage ratio. b = Method-B correction factor. n/a: no usable baseline measure (take-up and the two training outcomes have none; the mental-health and social-skills baseline variables are degenerate placeholders).

S4 Mental health and income measurement

Mental-health results do not establish broad average deterioration. GHQ-8 and alternative scores are retained with item-level multiplicity corrections and exploratory self-worth analysis. These are follow-up-only self-reports; participation and norm-tercile patterns do not identify mechanisms. Monitoring is precautionary, not an estimated welfare effect.

Table S33: Mental health: incremental effects on the GHQ-8 index, randomization inference, and attrition bounds

	T1	T2	T3	RI p (T3)	Holm-9 p (T3)
<i>Panel A: ITT under the published PDS-LASSO specification</i>					
GHQ-8 index (higher = better)	0.036 (0.160)	-0.140 (0.180)	-0.076 (0.149)	0.554	1.000
<i>Panel B: marginal Lee (2009) bounds for the GHQ-8 index</i>					
T1	[-0.052, 0.120]			trim 0.068 (control)	
T2	[-0.145, -0.138]			trim 0.002 (treated)	
T3	[-0.279, 0.009]			trim 0.090 (treated)	

Notes: The index is the GHQ-8, built from the follow-up GHQ items with every item coded so that a higher score indicates better mental health, and z-scored by the control group. Item reversal is built into the GHQ response wording, so no subset reverse-coding is applied. Cronbach's $\alpha = 0.81$ with every item-rest correlation positive, and the index correlates +0.33 with the standalone happiness item. Three alternative measures — the full GHQ-12 Likert, the GHQ caseness count, and the two-item self-worth subindex — are reported in Appendix Table S35. Panel A: coefficients on arm indicators from the published PDS-LASSO specification (post-double-selection over the 65 candidate baseline controls, strata fixed effects partialled, village-clustered SEs; no baseline mental-health measure exists). RI p : randomization inference with 1,000 within-strata permutations of the four-category village assignment (arm counts fixed), post-double-selection controls held fixed, $|t|$ criterion. Holm-9 p : Holm step-down across the nine-main-outcome family within arm, combining this table's mental-health p -value with the other outcomes' p -values from the reimplemented PDS specification of Table S20; because that reimplementaion can differ modestly from the published estimates of the main-paper ITT table, the correction is approximate (the qualitative conclusion—Holm $p = 1.0$ —is insensitive to which set is used). Panel B: manual marginal Lee bounds; the group with the higher follow-up response rate is trimmed from above and below by the response-rate differential. Sample: respondents with complete item data. The canonical Lee bounds cited in the text are those of the main-outcome Lee table (Table S16, T3 interval $[-0.28, 0.08]$), computed by the estimation pipeline on its outcome sample; the small differences between the two panels reflect the sample definitions, not the method. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Mental-health items and scoring

Table S34: Mental health: item-level treatment effects

	Safe Spaces (T1)		+ Training (T2)		+ Announced Grant (T3)	
	ITT	Holm p	ITT	Holm p	ITT	Holm p
<i>Panel A: individual items (control-SD units; higher = better)</i>						
Felt playing a useful role in life	0.056	1.000	-0.048	1.000	0.012	1.000
Felt could not overcome difficulties	0.038	1.000	-0.027	1.000	-0.042	1.000
Able to enjoy normal daily activities	0.064	1.000	0.029	1.000	-0.022	1.000
Able to face own problems	0.054	1.000	-0.072	1.000	-0.018	1.000
Felt unhappy and depressed	0.079	1.000	-0.083	1.000	-0.025	1.000
Lost self-confidence	-0.020	1.000	-0.187	1.000	-0.253**	0.252
Considered self worthless	-0.047	1.000	-0.197	1.000	-0.277**	0.156
Felt reasonably happy, all considered	0.122	1.000	0.061	1.000	0.085	1.000
<i>Panel B: GHQ-8 index (control-SD units)</i>						
GHQ-8 index	0.065	(0.157)	-0.113	(0.168)	-0.116	(0.147)

Notes: Each cell is the coefficient on the arm indicator from a regression of the item (Panel A) or index (Panel B) on the three arm indicators, strata fixed effects, and age dummies, with village-clustered standard errors (Panel B reports SEs in the adjacent column). All items are GHQ-style with four response categories referenced to 'as usual'. Every item is oriented so that higher values indicate better mental health: the GHQ builds the reversal into the item-specific response scales, so the items require no subset reverse-coding and none is applied here. Item coefficients are in control-group standard-deviation units. Holm p -values adjust across the eight items within arm. Panel B reports the summed GHQ-8 index in this specification; the PDS-LASSO estimates with PDS-LASSO controls are in Table S33. The module was administered at follow-up only. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The concentration in two self-worth items does not survive item-level Holm correction. This GHQ-8 row uses strata and age controls; the preceding table uses PDS-LASSO. The specifications and samples are not identical. The registration crosswalk classifies mental health as post-registration because the registration does not name the domain.

Alternative scores and exploratory welfare patterns

Table S35: Alternative mental-health measures

	T1: Safe Spaces	T2: SS + Training	T3: SS+Tr.+ Ann. Grant
Full GHQ-12 Likert (higher = better)	0.042 (0.161)	-0.103 (0.184)	-0.055 (0.158)
<i>RI p-val</i>	[0.671]	[0.672]	[0.647]
GHQ caseness count (0–8; higher = worse)	-0.043 (0.269)	0.281 (0.285)	0.243 (0.240)
<i>RI p-val</i>	[0.770]	[0.633]	[0.361]
Self-worth subindex (2 items; higher = better)	-0.037 (0.146)	-0.258 (0.157)	-0.243* (0.132)
<i>RI p-val</i>	[0.825]	[0.176]	[0.054]

Notes: Alternative constructions of the follow-up GHQ module, reported as robustness for the GHQ-8 index of Table S33. The GHQ-12 Likert and the self-worth subindex are z-scored by the control group and oriented so that a higher score is better mental health; **the caseness count is oriented the other way** — it is the raw 0–8 count of items answered in the two worst response categories, so a positive coefficient means worse mental health. The self-worth subindex comprises the two items (lost confidence, felt worthless) that carry most of the movement in the GHQ-8; it is a decomposition of the GHQ-8 index, not independent evidence. Coefficients are on arm indicators from the published PDS-LASSO specification (post-double-selection over the 65 candidate baseline controls, strata fixed effects partialled, village-clustered SEs in parentheses; no baseline mental-health measure exists). *RI p*: randomization inference with 1,000 within-strata permutations of the four-category village assignment (arm counts fixed), post-double-selection controls held fixed, $|t|$ criterion. No Holm correction is reported: these measures are not members of the nine-outcome main family over which that correction is defined. Sample as in Table S33. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table S36: Mental-health measures: exposure, attitude coupling, and norms gradient

	GHQ-8 index	Self-worth subindex
T3 non-participants vs. control	-0.177 (0.136)	-0.330*** (0.117)
T3 participants vs. control	-0.013 (0.165)	-0.245* (0.145)
Attitude-change coupling, all T3	0.001 (0.076)	-0.042 (0.072)
Attitude-change coupling, T3 participants	0.170* (0.098)	0.079 (0.095)
T3 ITT, norms tercile 1 (most conservative)	-0.756** (0.357)	-0.751** (0.279)
T3 ITT, norms tercile 2	0.054 (0.241)	-0.094 (0.202)
T3 ITT, norms tercile 3 (least conservative)	-0.173 (0.259)	-0.411 (0.289)

Notes: Both outcome columns are z-scored by the control group, higher = better. Rows 1–2: coefficients on indicators for T3 non-participants and T3 participants (relative to control girls) from one regression on the T3-and-control sample with strata fixed effects and age dummies, village-clustered SEs; participation is safe-space take-up, which is itself a choice, so the split is descriptive. Rows 3–4: coefficient on the within-person change in the indicator for naming any early-marriage disadvantage, which is binary at both waves, among T3 girls (row 3) and T3 participants (row 4), same controls. Rows 5–7: T3 coefficients from separate regressions within terciles of the baseline village social-norms (Unrestricted) index on the T3-and-control sample. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Income-generation components and baseline context

Table S37: IGA index components: incremental effects and infrastructure gradient

	Control mean	T1 ITT	T2 ITT	T3 ITT	T3×Econ. Infra. interaction
Any paid activity, last 30 days (entry)	0.525	0.124	-0.102	0.124	-0.240* (0.126)
Cash profits, IHS (income)	1.930	0.097	-0.033	-0.017	-0.116 (0.122)
No. of own-initiative activities (agency)	0.284	0.309*	0.152	0.094	-0.237* (0.122)
No. of activities, last 6 months (diversification)	0.539	0.172	0.073	0.078	-0.156 (0.109)
IGA index (benchmark)					-0.248** (0.111)

Notes: Components of the Anderson (2008) IGA index, each standardized by the control-group SD. Control mean is in the component's natural units (share, IHS francs, counts). ITT columns: coefficients on arm indicators from one regression per component with baseline IGA index (plus missingness indicator), strata fixed effects, and age dummies, village-clustered SEs. Final column: coefficient on T3×standardized economic-infrastructure index from the design-based interaction specification of the main-paper exploratory-heterogeneity section (T3 and control villages, strata fixed effects), with SE in parentheses; the benchmark row reports the same interaction for the full index. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table S38: Baseline IGA activity by village economic-infrastructure tercile

	Tercile 1 (weakest)	Tercile 2	Tercile 3 (strongest)	p (T3 vs. T1)
Any paid activity, last 30 days	0.057	0.079	0.065	0.704
Cash profits, last 30 days (FCFA)	3026.494	1284.356	2518.710	0.531
No. of activities, last 6 months	0.144	0.197	0.135	0.814
IGA index (control-normalized)	-0.024	-0.156	-0.029	0.969
Villages	58	58	57	

Notes: Girl-level baseline means by tercile of the village economic-infrastructure index. Final column: p -value of the tercile-3 vs. tercile-1 difference from a girl-level regression on tercile indicators with village-clustered SEs. Cash profits are raw FCFA (untrimmed); the IGA index is the Anderson (2008) baseline index normalized by the control group.

S5 Parametric heterogeneity

Tertile tables retain assignment coefficients and interactions with their standard errors; additive sums and repeated means are omitted. First-tertile effects plus interactions give other tertile effects. Stars use unadjusted analytic p -values, not familywise evidence. Moderators are not randomized.

Table S39: Main Outcomes - Heterogeneous Effects by Social Norms

	(1) Never mar.	(2) Never mar1	(3) Never preg.	(4) Not preg.	(5) No emo. viol.	(6) IGA	(7) Mental health
Safe spaces	0.038 [0.028]	0.031 [0.049]	0.159* [0.076]	0.029 [0.024]	-0.019 [0.039]	0.107 [0.248]	0.180 [0.219]
SS & Tertile 2	-0.003 [0.026]	0.013 [0.060]	-0.089 [0.110]	0.002 [0.033]	-0.065 [0.061]	-0.159 [0.230]	-0.079 [0.244]
SS & Tertile 3	-0.111* [0.046]	0.005 [0.086]	-0.327** [0.114]	-0.020 [0.034]	0.041 [0.077]	0.272 [0.366]	-0.138 [0.337]
Safe spaces+	0.050 [0.035]	0.055 [0.075]	0.091 [0.063]	0.009 [0.023]	-0.011 [0.033]	0.109 [0.204]	-0.152 [0.315]
SS & Tertile 2	0.022 [0.036]	-0.046 [0.082]	0.106 [0.079]	0.018 [0.021]	-0.094 [0.065]	-0.058 [0.217]	0.101 [0.352]
SS & Tertile 3	-0.185** [0.067]	-0.091 [0.119]	-0.243* [0.110]	-0.014 [0.037]	-0.003 [0.073]	-0.419 [0.291]	-0.054 [0.402]
SS + Train. + Ann. Grant	-0.000 [0.044]	-0.082 [0.061]	0.036 [0.070]	-0.005 [0.025]	-0.033 [0.033]	0.152 [0.198]	-0.262 [0.212]
Full & Tertile 2	0.042 [0.041]	0.166* [0.068]	0.075 [0.070]	0.001 [0.032]	-0.081 [0.065]	0.282 [0.259]	0.333 [0.288]
Full & Tertile 3	-0.059 [0.059]	0.125 [0.103]	-0.139 [0.116]	0.021 [0.036]	0.043 [0.059]	-0.467 [0.289]	0.220 [0.326]

Village-level clustered standard errors in brackets

* $p \leq 0.10$, ** $p \leq 0.05$, *** $p \leq 0.01$

Treatment effects interacted with tertiles of village-level social norms (1=most restrictive, 3=least restrictive)

Dependent variables: never been married (1), never been married if single at baseline (2), never been pregnant (3), not currently pregnant (4), no emotional violence (5), income generating activities (6), mental health (7)

All regressions control for strata (province) fixed effects, and baseline outcome when available. Other covariates are selected by post-double-selection lasso from the candidate list described in the main empirical-strategy section

SRH access: the signed prediction and its limits

Table S40: Main Outcomes - Heterogeneous Effects by Access to Reproductive Health Services

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Never mar.	Never mar1	Never preg.	Not preg.	No emo. viol.	IGA	Mental health
Safe spaces	0.024 [0.027]	0.126* [0.056]	-0.037 [0.056]	0.032 [0.020]	0.012 [0.038]	0.003 [0.225]	0.181 [0.179]
SS & Tertile 2	0.004 [0.031]	-0.173* [0.077]	0.228* [0.103]	-0.031 [0.029]	-0.056 [0.054]	0.133 [0.260]	0.114 [0.291]
SS & Tertile 3	-0.072 [0.040]	-0.130* [0.059]	-0.004 [0.118]	-0.003 [0.031]	-0.096 [0.079]	0.666 [0.419]	-0.367 [0.349]
Safe spaces+	0.003 [0.049]	0.066 [0.084]	0.057 [0.073]	0.025 [0.027]	-0.054 [0.061]	-0.336* [0.170]	0.078 [0.360]
SS & Tertile 2	-0.031 [0.062]	-0.152 [0.116]	-0.063 [0.081]	-0.037 [0.026]	0.026 [0.069]	0.254 [0.198]	-0.432 [0.394]
SS & Tertile 3	0.027 [0.064]	-0.060 [0.088]	0.030 [0.146]	0.006 [0.037]	-0.031 [0.085]	0.845** [0.316]	-0.033 [0.442]
SS + Train. + Ann. Grant	-0.005 [0.032]	0.037 [0.059]	-0.017 [0.052]	-0.008 [0.023]	-0.111 [0.062]	0.072 [0.197]	-0.202 [0.200]
Full & Tertile 2	-0.001 [0.038]	-0.083 [0.107]	-0.010 [0.071]	-0.040 [0.032]	0.117 [0.062]	-0.072 [0.257]	0.405 [0.283]
Full & Tertile 3	0.008 [0.051]	-0.029 [0.061]	0.107 [0.123]	0.054 [0.031]	0.056 [0.077]	0.384 [0.334]	-0.043 [0.328]
Constant	0.303*** [0.042]	0.585*** [0.058]	0.475*** [0.058]	0.926*** [0.028]	0.916*** [0.028]	-0.079 [0.121]	-0.003 [0.114]

Village-level clustered standard errors in brackets

* $p \leq 0.10$, ** $p \leq 0.05$, *** $p \leq 0.01$

Treatment effects interacted with tertiles of village-level access to reproductive health services (1=least accessible, 3=most accessible)

Dependent variables: never been married (1), never been married if single at baseline (2), never been pregnant (3), not currently pregnant (4), no emotional violence (5), income generating activities (6), mental health (7)

All regressions control for strata (province) fixed effects, and baseline outcome when available. Other covariates are selected by post-double-selection lasso from the candidate list described in the main empirical-strategy section

The model predicts larger marriage/fertility delays where SRH access is greater. No pregnancy/fertility interaction reaches 5% even before correction; the marginal middle-tertile cell is not monotone. The signed prediction is unsupported on that family. The initially unmarried subgroup is a separate exploratory comparison. Raw stars do not change that conclusion.

Table S41 reports the same tertile decomposition for the intermediate outcomes. The knowledge margin is the one that moves: safe spaces raise sexual and reproductive health knowledge most in the least-served villages.

Table S41: Intermediary Outcomes - Heterogeneous Effects by Access to Reproductive Health Services

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
	Finance	Gender att	Aspirations	SRH know	SRH bv.	MHM	Dec. mkg	Dec. mkg1	Hh head op.	Soc skills	Fam: Mar./Fert.	Fam: Econ.	Fam: Knowl.	Fam: Pref./Att.	Fam: Well-being	Fam: Summary
Safe spaces	-0.059 [0.167]	0.123 [0.194]	-0.302* [0.137]	0.417* [0.180]	0.262 [0.136]	0.020 [0.154]	0.184 [0.156]	0.236 [0.175]	-0.198 [0.231]	-0.090 [0.222]	0.073 [0.083]	-0.059 [0.188]	0.282 [0.171]	-0.142 [0.165]	0.019 [0.184]	0.080 [0.152]
SS & Tertile 2	-0.080 [0.198]	-0.333 [0.196]	-0.016 [0.115]	-0.635** [0.225]	-0.326* [0.133]	-0.021 [0.147]	-0.176 [0.290]	-0.086 [0.408]	0.138 [0.232]	0.280 [0.279]	0.076 [0.120]	0.047 [0.230]	-0.431* [0.216]	-0.216 [0.148]	0.104 [0.287]	-0.114 [0.236]
SS & Tertile 3	0.405 [0.256]	-0.014 [0.277]	0.295 [0.305]	-0.396 [0.436]	-0.320 [0.386]	-0.144 [0.300]	-0.269 [0.345]	-0.170 [0.435]	0.112 [0.308]	-0.188 [0.314]	-0.138 [0.118]	0.710* [0.315]	-0.303 [0.424]	0.217 [0.312]	-0.400 [0.333]	-0.069 [0.274]
Safe spaces+	-0.258* [0.122]	0.472** [0.162]	-0.081 [0.151]	-0.191 [0.210]	0.104 [0.156]	-0.022 [0.149]	0.180 [0.189]	0.395 [0.292]	0.454* [0.181]	0.067 [0.238]	0.040 [0.113]	-0.388** [0.136]	-0.258 [0.187]	0.230 [0.139]	-0.035 [0.243]	-0.064 [0.147]
SS & Tertile 2	0.193 [0.171]	-0.227 [0.162]	0.260 [0.153]	0.497** [0.190]	0.110 [0.126]	0.109 [0.134]	-0.203 [0.231]	-0.231 [0.334]	-0.673** [0.219]	0.037 [0.269]	-0.122 [0.138]	0.272 [0.169]	0.461** [0.174]	0.042 [0.147]	-0.104 [0.285]	0.113 [0.183]
SS & Tertile 3	0.849** [0.267]	-0.195 [0.299]	0.267 [0.343]	0.224 [0.423]	0.040 [0.342]	-0.441 [0.302]	-0.234 [0.341]	-0.328 [0.425]	-0.925** [0.323]	-0.205 [0.368]	0.050 [0.163]	1.164*** [0.294]	0.217 [0.381]	0.064 [0.349]	-0.198 [0.380]	0.295 [0.260]
SS + Train. + Ann. Grant	0.208 [0.139]	0.511*** [0.147]	0.121 [0.141]	0.226 [0.172]	0.093 [0.138]	0.025 [0.128]	-0.093 [0.170]	0.173 [0.190]	0.161 [0.211]	-0.212 [0.196]	0.029 [0.089]	0.160 [0.173]	0.201 [0.158]	0.400** [0.150]	-0.343 [0.200]	0.101 [0.137]
Full & Tertile 2	-0.153 [0.160]	0.143 [0.227]	0.065 [0.188]	0.215 [0.167]	0.210 [0.107]	0.254 [0.149]	0.071 [0.297]	-0.396 [0.389]	-0.484 [0.346]	0.753** [0.256]	-0.081 [0.120]	-0.135 [0.231]	0.179 [0.164]	0.126 [0.235]	0.731** [0.258]	0.370* [0.170]
Full & Tertile 3	-0.033 [0.224]	-0.345 [0.238]	-0.003 [0.328]	-0.314 [0.383]	0.000 [0.317]	-0.296 [0.275]	-0.091 [0.328]	-0.195 [0.343]	-0.337 [0.289]	-0.159 [0.341]	0.051 [0.139]	0.280 [0.262]	-0.391 [0.352]	-0.203 [0.321]	-0.052 [0.316]	-0.152 [0.280]
Constant	0.083 [0.097]	-0.230* [0.113]	0.020 [0.117]	-0.009 [0.125]	0.001 [0.106]	0.482*** [0.087]	0.027 [0.119]	-0.018 [0.108]	-0.050 [0.143]	-0.005 [0.158]	0.627*** [0.106]	-0.073 [0.102]	0.020 [0.105]	-0.216 [0.119]	-0.014 [0.133]	0.223 [0.114]
Control Mean T1	-0.012	0.107	-0.001	0.092	0.130	0.506	-0.027	0.093	0.013	0.044	0.054	0.075	0.075	0.063	0.156	0.329
Control Mean T2	0.116	-0.148	0.045	-0.080	-0.150	0.475	0.037	-0.088	-0.097	-0.013	0.181	0.092	-0.064	-0.057	-0.177	0.169
Control Mean T3	-0.162	-0.001	-0.075	-0.068	-0.057	0.734	0.029	-0.036	0.148	-0.020	0.282	-0.340	-0.003	-0.054	0.001	0.189
T1+Tertile 2	-0.138	-0.210	-0.318	-0.218	-0.064	-0.001	0.008	0.150	-0.060	0.190	0.149	-0.012	-0.149	-0.358	0.124	-0.034
p-value	0.386	0.153	0.027	0.324	0.659	0.996	0.977	0.704	0.754	0.488	0.168	0.947	0.447	0.012	0.671	0.876
T1+Tertile 3	0.346	0.110	-0.007	0.020	-0.058	-0.124	-0.085	0.067	-0.086	-0.278	-0.065	0.651	-0.021	0.075	-0.380	0.010
p-value	0.070	0.560	0.980	0.957	0.866	0.623	0.776	0.873	0.660	0.207	0.417	0.008	0.955	0.775	0.163	0.963
T2+Tertile 2	-0.065	0.245	0.179	0.305	0.214	0.087	-0.023	0.164	-0.219	0.104	-0.083	-0.117	0.203	0.272	-0.138	0.049
p-value	0.694	0.106	0.287	0.041	0.051	0.414	0.919	0.486	0.353	0.689	0.404	0.456	0.108	0.098	0.547	0.765
T2+Tertile 3	0.591	0.277	0.186	0.033	0.144	-0.463	-0.053	0.067	-0.470	-0.137	0.089	0.775	-0.041	0.295	-0.232	0.232
p-value	0.012	0.260	0.538	0.928	0.630	0.074	0.835	0.793	0.069	0.618	0.409	0.003	0.901	0.347	0.429	0.261
T3+Tertile 2	0.055	0.654	0.186	0.441	0.303	0.278	-0.021	-0.223	-0.323	0.541	-0.052	0.025	0.380	0.526	0.388	0.471
p-value	0.714	0.005	0.360	0.010	0.021	0.070	0.943	0.551	0.337	0.057	0.641	0.902	0.017	0.029	0.128	0.006
T3+Tertile 3	0.175	0.166	0.118	-0.088	0.093	-0.271	-0.183	-0.022	-0.176	-0.371	0.080	0.440	-0.190	0.197	-0.395	-0.052
p-value	0.310	0.357	0.688	0.797	0.740	0.255	0.495	0.935	0.374	0.176	0.436	0.033	0.548	0.478	0.109	0.833

Village-level clustered standard errors in brackets

* p ≤ 0.10 , ** p ≤ 0.05, *** p ≤ 0.01

Treatment effects interacted with tertiles of village-level access to reproductive health services (1=least accessible, 3=most accessible)

Dependent variables: financial services (1), gender attitudes (2), aspirations (3), reproductive health: knowledge (4), reproductive health: behavior (5), menstrual hygiene management (6), decision making power (7), decision making power in couple (8), household head opinion on gender norms (9), societal skills (10)

All regressions control for strata (province) fixed effects, and baseline outcome when available. Other covariates are selected by post-double-selection lasso from the candidate list described in Online Appendix S1

Table S42 reports heterogeneity across tertiles of transportation access, which yields no coherent pattern, consistent with the ambiguous sign the framework assigns to this comparative static.

Table S42: Main Outcomes - Heterogeneous Effects by Access to Public Transportation

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Never mar.	Never mar1	Never preg.	Not preg.	Emo. viol	IGA	Mental health
Safe spaces	0.000 [0.033]	0.132 [0.071]	-0.000 [0.082]	0.007 [0.020]	-0.045 [0.036]	0.243 [0.281]	0.096 [0.179]
SS & Tertile 2	0.003 [0.041]	-0.170 [0.088]	0.049 [0.106]	0.014 [0.031]	0.023 [0.068]	-0.101 [0.361]	-0.029 [0.297]
SS & Tertile 3	-0.009 [0.029]	-0.077 [0.064]	0.022 [0.091]	0.046 [0.025]	0.022 [0.062]	0.053 [0.354]	-0.084 [0.417]
Safe spaces+	-0.037 [0.057]	-0.012 [0.115]	0.043 [0.082]	-0.011 [0.030]	-0.146** [0.054]	0.110 [0.253]	-0.189 [0.327]
SS & Tertile 2	0.058 [0.068]	-0.030 [0.122]	0.024 [0.104]	0.038 [0.037]	0.149* [0.070]	-0.263 [0.299]	-0.085 [0.388]
SS & Tertile 3	0.047 [0.057]	0.139 [0.122]	0.005 [0.101]	0.001 [0.030]	0.099 [0.083]	0.032 [0.351]	0.613 [0.547]
SS + Train. + Ann. Grant	0.013 [0.028]	0.126* [0.058]	0.029 [0.066]	-0.006 [0.023]	-0.136 [0.074]	0.576** [0.218]	-0.019 [0.178]
Full & Tertile 2	-0.003 [0.042]	-0.205* [0.080]	0.048 [0.090]	0.019 [0.036]	0.170* [0.083]	-0.818** [0.284]	-0.219 [0.294]
Full & Tertile 3	-0.066 [0.050]	-0.099 [0.089]	-0.129 [0.086]	-0.009 [0.036]	0.006 [0.078]	-0.312 [0.303]	0.298 [0.304]
Control Mean T1	0.184	0.481	0.401	0.942	0.940	0.199	-0.200
Control Mean T2	0.189	0.480	0.329	0.906	0.877	0.151	0.145
Control Mean T3	0.253	0.578	0.409	0.896	0.959	-0.634	-0.018
T1+Tertile 2	0.003	-0.038	0.048	0.021	-0.022	0.142	0.067
p-value	0.900	0.455	0.460	0.399	0.698	0.522	0.777
T1+Tertile 3	-0.008	0.055	0.022	0.053	-0.023	0.296	0.012
p-value	0.760	0.293	0.772	0.005	0.656	0.331	0.976
T2+Tertile 2	0.021	-0.043	0.066	0.026	0.003	-0.154	-0.275
p-value	0.550	0.439	0.322	0.274	0.950	0.338	0.224
T2+Tertile 3	0.010	0.127	0.048	-0.010	-0.048	0.142	0.424
p-value	0.720	0.061	0.555	0.543	0.459	0.661	0.331
T3+Tertile 2	0.010	-0.079	0.077	0.013	0.033	-0.242	-0.237
p-value	0.749	0.151	0.209	0.645	0.324	0.196	0.325
T3+Tertile 3	-0.053	0.027	-0.100	-0.016	-0.131	0.263	0.280
p-value	0.300	0.773	0.223	0.612	0.000	0.383	0.312

Village-level clustered standard errors in brackets

* $p \leq 0.10$, ** $p \leq 0.05$, *** $p \leq 0.01$

Treatment effects interacted with tertiles of village-level access to transportation (1=least accessible, 3=most accessible)

Dependent variables: never been married (1), never been married if single at baseline (2), never been pregnant (3), not currently pregnant (4), experienced any emotional violence (5), income generating activities (6), mental health (7)

All regressions control for strata (province) fixed effects, and baseline outcome when available. Other covariates are selected by post-double-selection lasso from the candidate list described in Online Appendix S1

S6 Exploratory heterogeneity: methods and audit

GML targets. For each arm-versus-control comparison, let D denote assignment and Z baseline covariates. The baseline response is $b_0(Z) = E[Y(0) | Z]$ and the treatment response $s_0(Z) = E[Y(1) - Y(0) | Z]$. Auxiliary samples train proxies $B(Z)$ and $S(Z)$; held-out samples estimate features of the response, not individual causal effects. The four learners are random forest, gradient boosting, elastic net, and partial least squares. The analysis uses 200 repeated splits with median aggregation. BLP p -values are doubled; median aggregation of 95% split-level GATES/CLAN bounds gives 90% coverage under the stated dominant-learner assumption.

With assignment probability $p(Z)$ and $w(Z) = \{p(Z)[1 - p(Z)]\}^{-1}$, the BLP uses the weighted regression

$$Y = \alpha'X_1 + \beta_1[D - p(Z)] + \beta_2[D - p(Z)][S(Z) - \bar{S}] + \varepsilon,$$

where X_1 contains baseline controls and $B(Z)$. Nonzero β_2 indicates predictive heterogeneity; $\beta_2 = 1$ alone does not establish perfect prediction.

For proxy quintiles G_k , GATES estimates $\gamma_k = E[s_0(Z) | G_k]$ using $[D - p(Z)]1\{G_k\}$ interactions. CLAN compares characteristics $E[Z | G_5] - E[Z | G_1]$, a descriptive contrast, not an effect of the characteristic. Learners are compared using predictive heterogeneity criteria, including weighted squared GATES estimates $\sum_k \hat{\gamma}_k^2 \hat{P}(G_k)$. Extreme-quintile regions aggregate split-level bounds as described in Table S43.

Design-based interactions. The primary participation-pattern audit uses assignment-by-moderator interactions, strata effects, and village-clustered standard errors without a machine-learning first stage. The focal family is T3 crossed with norms/economic infrastructure and take-up/never pregnant/IGA. Selected after registration and exploration, it is post-hoc: its smallest Holm-RI p is 0.256. The joint sign-pattern search test gives $p = 0.103$ within T3 and $p = 0.260$ across all three arms (the main-paper focal-interaction table and discussion).

Interpretation. Scores/decompositions cannot identify returns to participation or effects of externally increasing attendance. Choice and disrupted provision both enter non-participation. The income pattern also admits substitution for absent services. The following tables retain the selection and delivery checks preventing an established barrier interpretation. Detailed learner, full-quintile, covariate-by-covariate CLAN, and repeated decomposition outputs remain in the full technical supplement.

Method reference. Chernozhukov, V., M. Demirer, E. Duflo, and I. Fernández-Val (2025), “Fisher–Schultz Lecture: Generic Machine Learning Inference on Heterogeneous Treatment Effects in Randomized Experiments, With an Application to Immunization in India,” *Econometrica* 93(4): 1121–1164. Full implementation details are in the replication scripts.

Extreme-quintile magnitudes

Table S43: GATES extreme-quintile estimates for the outcomes discussed in the text

Outcome	Arm	G1 (least affected)	G5 (most affected)
Prog. take-up	T1	0.010 (−0.073, 0.091)	0.529 (0.397, 0.657)
Prog. take-up	T2	0.059 (−0.041, 0.161)	0.342 (0.206, 0.485)
Prog. take-up	T3	0.035 (−0.046, 0.112)	0.680 (0.558, 0.801)
Never Pregnant	T1	−0.167 (−0.319, −0.022)	0.325 (0.120, 0.527)
Never Pregnant	T2	−0.157 (−0.307, −0.006)	0.417 (0.261, 0.576)
Never Pregnant	T3	−0.182 (−0.316, −0.055)	0.282 (0.139, 0.425)
IGA	T3	−0.952 (−1.328, −0.576)	1.255 (0.792, 1.695)
No Emo. Viol.	T3	−0.205 (−0.378, −0.026)	0.050 (−0.106, 0.198)
Mental Health	T3	−0.874 (−1.300, −0.444)	0.756 (0.295, 1.211)

Notes: Sorted group average treatment effects (GATES) for the bottom (G1) and top (G5) quintiles of the ML proxy score, median-aggregated across the 200 sample splits of the single estimation run described in the GML methods section; 90 percent coverage regions in parentheses (median-aggregated 95 percent bounds). Binary outcomes are in the desirable orientation (Never Pregnant; No Emo. Viol.), so G1–G5 contrasts read as gains; take-up and the binary outcomes are in proportion units (e.g., the T3 take-up G5 effect of 0.680 is a 68 percentage-point increase), and IGA and Mental Health are in control-group standard deviations. Full profiles are retained in the archived technical supplement; this submission option reports the extreme-quintile table only.

These are group-average assignment effects after ranking by prediction scores, not effects of targeting groups in a new intervention. Large contrasts do not override multiplicity, selection, or prediction uncertainty. These selected outcomes illustrate the secondary heterogeneity analysis; they are not the full universe of GML results.

GML estimates: BLP, GATES, and CLAN

The tables and figure below report the generic machine learning estimates summarized in the paper: the best linear predictor of heterogeneity, the sorted group average treatment effects, and the classification analysis. All come from a single estimation run of 200 splits per arm with median aggregation across splits.

Table S44: BLP Results: Main Outcomes

Outcome	Arm	ATE	HTE
Never Married	T1	−0.004 [1.000]	0.439 [0.242]
Never Married	T2	−0.005 [1.000]	0.689*** [0.004]
Never Married	T3	0.013 [1.000]	0.408 [0.186]
Never Married	T4	0.012 [1.000]	0.441 [0.187]
Age at Mar.	T1	0.209 [0.269]	0.149 [0.766]
Age at Mar.	T2	0.079 [1.000]	0.431 [0.131]
Age at Mar.	T3	0.139 [0.621]	0.244 [0.519]
Age at Mar.	T4	0.073 [1.000]	0.200 [0.248]
Mar. bf18	T1	0.004 [1.000]	0.246 [0.555]
Mar. bf18	T2	−0.026 [0.901]	0.269 [0.289]
Mar. bf18	T3	0.028 [0.842]	0.146 [0.896]
Mar. bf18	T4	−0.015 [1.000]	0.145 [0.897]
Never Pregnant	T1	0.029 [0.609]	0.911*** [0.000]
Never Pregnant	T2	0.050 [0.188]	0.920*** [0.000]
Never Pregnant	T3	0.031 [0.584]	0.888*** [0.000]
Never Pregnant	T4	0.036 [0.349]	0.895*** [0.000]
Curr. Pregnant	T1	0.017 [0.737]	0.626 [0.408]
Curr. Pregnant	T2	0.005 [1.000]	0.472 [0.712]
Curr. Pregnant	T3	−0.001 [1.000]	0.511 [0.597]
Curr. Pregnant	T4	0.000 [1.000]	0.498 [0.591]
Preg. bf18	T1	−0.008 [1.000]	0.408 [0.179]
Preg. bf18	T2	0.001 [1.000]	0.449 [0.115]
Preg. bf18	T3	−0.007 [1.000]	0.278 [0.508]
Preg. bf18	T4	−0.004 [1.000]	0.275 [0.433]
Emo. Violence	T1	−0.049* [0.088]	0.800*** [0.004]
Emo. Violence	T2	−0.062** [0.017]	0.893*** [0.000]
Emo. Violence	T3	−0.053** [0.022]	0.880*** [0.000]
Emo. Violence	T4	−0.054** [0.013]	1.112*** [0.000]
IGA	T1	0.229*** [0.004]	0.982*** [0.000]
IGA	T2	−0.003 [1.000]	1.002*** [0.000]
IGA	T3	0.041 [1.000]	1.037*** [0.000]
IGA	T4	0.030 [1.000]	1.097*** [0.000]
Mental Health	T1	0.058 [0.718]	1.049*** [0.000]
Mental Health	T2	−0.081 [0.535]	1.002*** [0.000]
Mental Health	T3	−0.100 [0.261]	0.860*** [0.000]
Mental Health	T4	−0.106 [0.166]	0.908*** [0.000]

Notes: Columns report BLP components by arm. Estimates are obtained via the specification in equation C3. See Appendix S6 for GML methodology. ATE and HTE correspond to parameters β_1 and β_2 , respectively. HTE estimates significantly different from zero serve as evidence that heterogeneity is present in the true CATE, and that heterogeneity was detectable by ML. Estimates are median-aggregated over 200 random splits. p -values are median-aggregated over 200 splits and then multiplied by two, which allows for inference at usual significance levels for hypothesis tests. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. p -values are rounded to 3 decimal places. Arm T4 combines T2 and T3 as if they were one treatment. ML algorithms are selected according to the goodness-of-fit of C5. Selection criteria are stated in Section S6.

Table S45: BLP Results: Intermediate Outcomes

Outcome	Arm	ATE	HTE
Adv M. Early Nb	T1	−0.125 [0.108]	1.010*** [0.000]
Adv M. Early Nb	T2	−0.313*** [0.006]	0.923*** [0.000]
Adv M. Early Nb	T3	−0.443*** [0.000]	0.912*** [0.000]
Adv M. Early Nb	T4	−0.374*** [0.000]	1.018*** [0.000]
Marriage & Fertility	T1	0.041 [0.866]	0.393 [0.191]
Marriage & Fertility	T2	−0.005 [1.000]	0.727*** [0.002]
Marriage & Fertility	T3	−0.010 [1.000]	0.362** [0.029]
Marriage & Fertility	T4	0.003 [1.000]	0.401* [0.095]
Econ Activity & Finance	T1	0.141 [0.131]	1.012*** [0.000]
Econ Activity & Finance	T2	0.058 [0.891]	0.979*** [0.000]
Econ Activity & Finance	T3	0.108 [0.329]	0.975*** [0.000]
Econ Activity & Finance	T4	0.076 [0.494]	1.107*** [0.000]
Knowledge	T1	0.070 [0.488]	0.952*** [0.000]
Knowledge	T2	−0.001 [1.000]	0.957*** [0.000]
Knowledge	T3	0.128* [0.050]	0.994*** [0.000]
Knowledge	T4	0.035 [1.000]	0.974*** [0.000]
Preferences & Attitudes	T1	−0.114 [0.203]	0.834*** [0.000]
Preferences & Attitudes	T2	0.249*** [0.001]	0.870*** [0.000]
Preferences & Attitudes	T3	0.367*** [0.000]	1.008*** [0.000]
Preferences & Attitudes	T4	0.297*** [0.000]	0.959*** [0.000]
Wellbeing & Safety	T1	−0.091 [0.357]	1.028*** [0.000]
Wellbeing & Safety	T2	−0.163** [0.041]	0.941*** [0.000]
Wellbeing & Safety	T3	−0.133** [0.037]	0.961*** [0.000]
Wellbeing & Safety	T4	−0.167*** [0.006]	1.009*** [0.000]
Super-index	T1	−0.004 [1.000]	0.807*** [0.000]
Super-index	T2	−0.006 [1.000]	0.760*** [0.000]
Super-index	T3	0.148** [0.022]	0.848*** [0.000]
Super-index	T4	0.079 [0.320]	0.818*** [0.000]

Notes: Columns report BLP components by arm. Estimates are obtained via the specification in equation C3. See Appendix S6 for GML methodology. ATE and HTE correspond to parameters β_1 and β_2 , respectively. HTE estimates significantly different from zero serve as evidence that heterogeneity is present in the true CATE, and that heterogeneity was detectable by ML. Estimates are median-aggregated over 200 random splits. p -values are median-aggregated over 200 splits and then multiplied by two, which allows for inference at usual significance levels for hypothesis tests. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. p -values are rounded to 3 decimal places. Arm T4 combines T2 and T3 as if they were one treatment. ML algorithms are selected according to the goodness-of-fit of C5. Selection criteria are stated in Section S6.

Table S46: GATES Results: Main Outcomes

Outcome	Arm	G1	G5	G5-G1
Never Married	T1	−0.060 (−0.174, 0.052)	0.077 (−0.048, 0.204)	0.115 (−0.065, 0.282)
Never Married	T2	−0.127 (−0.254, 0.003)	0.110 (−0.012, 0.232)	0.232* (0.079, 0.382)
Never Married	T3	−0.055 (−0.177, 0.055)	0.084 (−0.026, 0.199)	0.132 (−0.021, 0.284)
Never Married	T4	−0.057 (−0.192, 0.070)	0.092 (−0.009, 0.197)	0.145 (−0.016, 0.302)
Age at Mar.	T1	−0.146 (−0.840, 0.551)	0.611 (−0.084, 1.301)	0.079 (−0.854, 0.996)
Age at Mar.	T2	−0.304 (−1.074, 0.451)	0.586 (−0.084, 1.263)	0.740 (−0.114, 1.569)
Age at Mar.	T3	−0.164 (−0.952, 0.652)	0.600 (−0.098, 1.298)	0.219 (−0.664, 1.093)
Age at Mar.	T4	−0.230 (−0.916, 0.483)	0.464 (−0.101, 1.033)	0.463 (−0.373, 1.276)
Mar. bf18	T1	−0.094 (−0.265, 0.074)	0.085 (−0.089, 0.268)	0.053 (−0.156, 0.265)
Mar. bf18	T2	−0.128 (−0.308, 0.046)	0.077 (−0.102, 0.256)	0.158 (−0.051, 0.375)
Mar. bf18	T3	−0.058 (−0.220, 0.109)	0.109 (−0.048, 0.270)	0.060 (−0.144, 0.247)
Mar. bf18	T4	−0.072 (−0.224, 0.073)	0.079 (−0.075, 0.230)	0.052 (−0.124, 0.234)
Never Pregnant	T1	−0.172* (−0.321, −0.023)	0.324* (0.125, 0.525)	0.493* (0.271, 0.717)
Never Pregnant	T2	−0.166* (−0.318, −0.012)	0.411* (0.253, 0.573)	0.569* (0.375, 0.771)
Never Pregnant	T3	−0.180* (−0.315, −0.046)	0.283* (0.140, 0.428)	0.458* (0.288, 0.631)
Never Pregnant	T4	−0.160* (−0.288, −0.030)	0.334* (0.192, 0.471)	0.485* (0.321, 0.652)
Curr. Pregnant	T1	−0.031 (−0.125, 0.056)	0.073 (−0.027, 0.178)	0.074 (−0.049, 0.197)
Curr. Pregnant	T2	−0.042 (−0.136, 0.051)	0.054 (−0.044, 0.152)	0.055 (−0.070, 0.177)
Curr. Pregnant	T3	−0.053 (−0.142, 0.030)	0.049 (−0.043, 0.141)	0.065 (−0.043, 0.175)
Curr. Pregnant	T4	−0.043 (−0.114, 0.027)	0.044 (−0.051, 0.137)	0.057 (−0.045, 0.156)
Preg. bf18	T1	−0.093 (−0.188, 0.000)	0.036 (−0.048, 0.125)	0.103 (−0.016, 0.226)
Preg. bf18	T2	−0.056 (−0.166, 0.038)	0.083 (−0.040, 0.210)	0.125 (−0.024, 0.278)
Preg. bf18	T3	−0.053 (−0.154, 0.042)	0.040 (−0.044, 0.138)	0.063 (−0.068, 0.192)
Preg. bf18	T4	−0.049 (−0.130, 0.023)	0.034 (−0.045, 0.121)	0.047 (−0.054, 0.149)
Emo. Violence	T1	−0.195* (−0.365, −0.030)	0.037 (−0.076, 0.153)	0.220* (0.055, 0.392)
Emo. Violence	T2	−0.263* (−0.409, −0.119)	0.138 (−0.026, 0.312)	0.410* (0.215, 0.593)
Emo. Violence	T3	−0.205* (−0.380, −0.029)	0.050 (−0.086, 0.195)	0.258* (0.063, 0.455)
Emo. Violence	T4	−0.259* (−0.385, −0.136)	0.118 (−0.043, 0.282)	0.386* (0.205, 0.559)
IGA	T1	−0.743* (−1.090, −0.404)	1.450* (0.842, 2.080)	2.199* (1.594, 2.827)
IGA	T2	−0.836* (−1.242, −0.430)	0.934* (0.463, 1.393)	1.780* (1.213, 2.320)
IGA	T3	−0.968* (−1.343, −0.601)	1.259* (0.809, 1.702)	2.215* (1.713, 2.719)
IGA	T4	−0.927* (−1.249, −0.601)	1.151* (0.796, 1.507)	2.071* (1.642, 2.500)
Mental Health	T1	−0.846* (−1.250, −0.436)	0.975* (0.593, 1.340)	1.830* (1.281, 2.343)
Mental Health	T2	−0.930* (−1.301, −0.555)	0.968* (0.587, 1.339)	1.892* (1.426, 2.345)
Mental Health	T3	−0.875* (−1.296, −0.450)	0.751* (0.293, 1.207)	1.629* (1.001, 2.243)
Mental Health	T4	−0.874* (−1.269, −0.479)	0.844* (0.469, 1.211)	1.721* (1.236, 2.211)

Notes: Sorted group average treatment effects (GATES) for the least affected (G1) and most affected (G5) quintiles of the ML proxy score, median-aggregated across 200 sample splits. The specification is equation C4; see Appendix S6 for GML methodology. 90 percent coverage regions are reported in parentheses. * indicates that the median-aggregated 90 percent confidence interval excludes zero. Outcomes are constructed such that a high value is more desirable (except Preg. bf18). The contrast G5–G1 is the difference in treatment effects between these groups. Arm T4 combines T2 and T3 as if they were one treatment. ML algorithms are selected according to the goodness-of-fit of C6. Selection criteria are stated in Section S6.

Table S47: GATES Results: Intermediate Outcomes

Outcome	Arm	G1	G5	G5-G1
Adv M. Early Nb	T1	-1.039* (-1.439, -0.651)	1.124* (0.647, 1.600)	2.172* (1.609, 2.752)
Adv M. Early Nb	T2	-1.036* (-1.430, -0.643)	0.542* (0.121, 0.963)	1.572* (1.086, 2.061)
Adv M. Early Nb	T3	-1.036* (-1.421, -0.636)	0.020 (-0.278, 0.329)	1.053* (0.613, 1.490)
Adv M. Early Nb	T4	-1.055* (-1.449, -0.641)	0.335* (0.031, 0.656)	1.396* (0.944, 1.847)
Marriage & Fertility	T1	-0.132 (-0.383, 0.117)	0.231 (-0.049, 0.525)	0.307 (-0.055, 0.654)
Marriage & Fertility	T2	-0.263 (-0.546, 0.008)	0.328 (-0.004, 0.661)	0.563* (0.198, 0.935)
Marriage & Fertility	T3	-0.228 (-0.519, 0.061)	0.164 (-0.086, 0.417)	0.347* (0.014, 0.667)
Marriage & Fertility	T4	-0.171 (-0.423, 0.077)	0.193 (-0.063, 0.463)	0.355* (0.033, 0.662)
Econ Activity & Finance	T1	-0.574* (-0.902, -0.234)	1.153* (0.717, 1.597)	1.726* (1.234, 2.217)
Econ Activity & Finance	T2	-0.607* (-0.985, -0.228)	0.900* (0.460, 1.362)	1.515* (0.986, 2.039)
Econ Activity & Finance	T3	-0.644* (-1.001, -0.279)	1.043* (0.646, 1.446)	1.692* (1.229, 2.148)
Econ Activity & Finance	T4	-0.698* (-1.041, -0.362)	1.047* (0.707, 1.389)	1.730* (1.306, 2.158)
Knowledge	T1	-0.902* (-1.284, -0.518)	1.072* (0.507, 1.638)	1.972* (1.365, 2.571)
Knowledge	T2	-0.685* (-0.978, -0.388)	0.813* (0.514, 1.113)	1.486* (1.116, 1.851)
Knowledge	T3	-0.638* (-0.944, -0.328)	0.716* (0.409, 1.048)	1.367* (0.982, 1.753)
Knowledge	T4	-0.741* (-0.981, -0.498)	0.706* (0.433, 0.986)	1.445* (1.131, 1.771)
Preferences & Attitudes	T1	-0.627* (-0.946, -0.315)	0.607* (0.168, 1.043)	1.211* (0.709, 1.724)
Preferences & Attitudes	T2	-0.543* (-0.977, -0.080)	0.739* (0.377, 1.100)	1.248* (0.738, 1.761)
Preferences & Attitudes	T3	-0.314* (-0.624, -0.017)	1.193* (0.829, 1.557)	1.492* (1.079, 1.915)
Preferences & Attitudes	T4	-0.457* (-0.789, -0.129)	0.980* (0.676, 1.307)	1.440* (1.033, 1.838)
Wellbeing & Safety	T1	-0.990* (-1.338, -0.637)	0.836* (0.453, 1.182)	1.808* (1.367, 2.260)
Wellbeing & Safety	T2	-1.143* (-1.542, -0.743)	0.958* (0.422, 1.490)	2.109* (1.448, 2.728)
Wellbeing & Safety	T3	-1.010* (-1.367, -0.654)	0.803* (0.344, 1.276)	1.808* (1.249, 2.355)
Wellbeing & Safety	T4	-1.093* (-1.407, -0.795)	0.845* (0.439, 1.248)	1.931* (1.459, 2.421)
Super-index	T1	-0.534* (-0.835, -0.245)	0.536* (0.188, 0.873)	1.068* (0.659, 1.465)
Super-index	T2	-0.637* (-0.975, -0.305)	0.482* (0.098, 0.863)	1.126* (0.666, 1.564)
Super-index	T3	-0.436* (-0.679, -0.181)	0.812* (0.517, 1.109)	1.239* (0.901, 1.582)
Super-index	T4	-0.487* (-0.752, -0.213)	0.644* (0.370, 0.919)	1.136* (0.790, 1.460)

Notes: Sorted group average treatment effects (GATES) for the least affected (G1) and most affected (G5) quintiles of the ML proxy score, median-aggregated across 200 sample splits. The specification is equation C4; see Appendix S6 for GML methodology. 90 percent coverage regions are reported in parentheses. * indicates that the median-aggregated 90 percent confidence interval excludes zero. Outcomes are constructed such that a high value is more desirable (except Preg. bf18). The contrast G5-G1 is the difference in treatment effects between these groups. Arm T4 combines T2 and T3 as if they were one treatment. ML algorithms are selected according to the goodness-of-fit of C6. Selection criteria are stated in Section S6.

Table S48: CLAN Estimates for T1 & T2: Main outcomes $\times$ Additional Village-level

Outcome	Characteristic	T1			T2		
		G1	G5	G5-G1	G1	G5	G5-G1
Never Married	Security	0.292	0.014	-0.251 (-0.542, 0.051)	0.319	0.152	-0.165 (-0.527, 0.208)
Never Married	Eligible Pop.	89.027	102.932	13.391 (-4.021, 31.688)	80.039	89.198	8.991 (-10.981, 30.064)
Never Married	Ethnic Frac.	0.215	0.219	0.003 (-0.069, 0.076)	0.249	0.225	-0.023 (-0.136, 0.086)
Never Married	Religious Frac.	0.370	0.361	-0.008 (-0.024, 0.009)	0.363	0.367	0.004 (-0.030, 0.038)
Age at Mar.	Security	0.239	0.184	-0.053 (-0.297, 0.172)	0.211	0.288	0.079 (-0.148, 0.309)
Age at Mar.	Eligible Pop.	94.891	92.410	-2.130 (-16.665, 11.860)	87.479	85.092	-2.130 (-14.961, 11.776)
Age at Mar.	Ethnic Frac.	0.229	0.219	-0.010 (-0.076, 0.056)	0.251	0.234	-0.016 (-0.094, 0.056)
Age at Mar.	Religious Frac.	0.363	0.367	0.005 (-0.013, 0.022)	0.375	0.361	-0.015 (-0.039, 0.010)
Mar. bf18	Security	0.232	0.271	0.050 (-0.153, 0.260)	0.123	0.380	0.253 (-0.028, 0.534)
Mar. bf18	Eligible Pop.	92.991	86.366	-7.861 (-19.429, 4.514)	88.235	83.503	-5.334 (-22.646, 12.183)
Mar. bf18	Ethnic Frac.	0.230	0.216	-0.017 (-0.078, 0.042)	0.240	0.241	0.003 (-0.079, 0.083)
Mar. bf18	Religious Frac.	0.366	0.367	0.002 (-0.015, 0.019)	0.377	0.357	-0.019 (-0.043, 0.003)
Never Pregnant	Security	0.235	-0.028	-0.257 (-0.663, 0.135)	0.091	0.234	0.134 (-0.336, 0.616)
Never Pregnant	Eligible Pop.	82.889	112.564	29.655* (6.575, 52.417)	77.353	94.289	16.170 (-5.463, 38.693)
Never Pregnant	Ethnic Frac.	0.208	0.209	0.003 (-0.101, 0.107)	0.224	0.217	-0.005 (-0.121, 0.114)
Never Pregnant	Religious Frac.	0.371	0.367	-0.004 (-0.027, 0.019)	0.380	0.369	-0.010 (-0.044, 0.023)
Curr. Pregnant	Security	0.186	0.220	0.020 (-0.229, 0.281)	0.245	0.292	0.027 (-0.192, 0.255)
Curr. Pregnant	Eligible Pop.	87.425	102.820	15.500 (-1.195, 32.455)	86.863	81.635	-5.754 (-18.820, 8.472)
Curr. Pregnant	Ethnic Frac.	0.235	0.187	-0.042 (-0.121, 0.029)	0.264	0.211	-0.050 (-0.125, 0.022)
Curr. Pregnant	Religious Frac.	0.367	0.361	-0.005 (-0.030, 0.018)	0.364	0.374	0.011 (-0.011, 0.033)

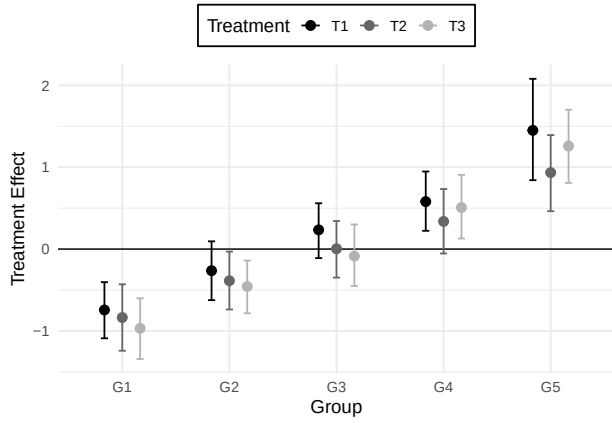
Notes: G1 and G5 report the mean of the indicated characteristic in the bottom and top quintiles of the ML proxy score. G5-G1 reports the corresponding contrast followed by its 90 percent confidence interval in parentheses. * indicates that the median-aggregated 90 percent confidence interval excludes zero. Outcomes and characteristics are constructed such that a high value is better (except Preg. bf18). As such, a *positive* contrast (G5-G1) means that the treatment had a greater effect for AGYWs in villages with *high* values of the characteristic at baseline. A *negative* contrast means treatment had a greater effect for AGYWs in villages with *low* values of the characteristic at baseline. Estimates and confidence bounds are median-aggregated across 200 sample splits. See Appendix S6 for the GML methodology. Arm T4 combines T2 and T3 as if they were one treatment. ML algorithms are selected according to the goodness-of-fit of C6. Selection criteria are stated in Section S6.

Table S49: CLAN Estimates for T3 & T4: Intermediate outcomes $\times$ Village-level

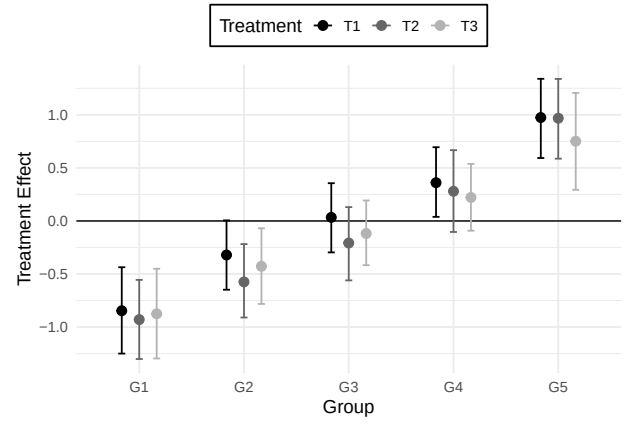
Outcome	Characteristic	T3			T4		
		G1	G5	G5-G1	G1	G5	G5-G1
Preferences & Attitudes	Social Norms	-0.410	0.326	0.712* (0.164, 1.280)	-0.287	0.201	0.494* (0.062, 0.936)
Preferences & Attitudes	SRH Access	0.284	0.067	-0.205 (-0.727, 0.306)	0.052	0.196	0.146 (-0.343, 0.635)
Preferences & Attitudes	Econ. Infrast.	0.325	0.163	-0.154 (-0.564, 0.247)	0.339	0.187	-0.156 (-0.484, 0.171)
Preferences & Attitudes	Male Exogamy	0.506	0.366	-0.139* (-0.243, -0.031)	0.491	0.358	-0.131* (-0.222, -0.040)
Preferences & Attitudes	Female Exogamy	0.330	0.330	0.002 (-0.204, 0.202)	0.377	0.296	-0.082 (-0.243, 0.081)
Preferences & Attitudes	Overall Exogamy	0.414	0.340	-0.073 (-0.206, 0.064)	0.429	0.320	-0.111 (-0.219, 0.000)
Wellbeing & Safety	Social Norms	-0.127	0.007	0.151 (-0.439, 0.726)	0.032	-0.029	-0.055 (-0.531, 0.408)
Wellbeing & Safety	SRH Access	-0.065	0.227	0.287 (-0.362, 0.950)	0.068	0.087	0.031 (-0.500, 0.566)
Wellbeing & Safety	Econ. Infrast.	0.129	0.253	0.123 (-0.394, 0.636)	0.109	0.245	0.139 (-0.268, 0.540)
Wellbeing & Safety	Male Exogamy	0.510	0.505	-0.002 (-0.123, 0.117)	0.502	0.493	-0.009 (-0.109, 0.091)
Wellbeing & Safety	Female Exogamy	0.408	0.243	-0.168 (-0.364, 0.029)	0.473	0.234	-0.242* (-0.395, -0.086)
Wellbeing & Safety	Overall Exogamy	0.451	0.404	-0.050 (-0.177, 0.078)	0.483	0.378	-0.106 (-0.216, 0.004)
Super-index	Social Norms	-0.422	0.359	0.776* (0.225, 1.332)	-0.161	0.061	0.226 (-0.163, 0.622)
Super-index	SRH Access	-0.019	0.346	0.372 (-0.205, 0.935)	-0.163	0.438	0.591* (0.169, 1.015)
Super-index	Econ. Infrast.	0.187	0.192	0.007 (-0.472, 0.467)	0.169	0.262	0.099 (-0.271, 0.449)
Super-index	Male Exogamy	0.513	0.384	-0.124 (-0.257, 0.002)	0.523	0.404	-0.117* (-0.228, -0.006)
Super-index	Female Exogamy	0.442	0.210	-0.232* (-0.423, -0.042)	0.530	0.178	-0.353* (-0.494, -0.211)
Super-index	Overall Exogamy	0.460	0.308	-0.151 (-0.307, 0.002)	0.510	0.298	-0.212* (-0.328, -0.093)

Notes: G1 and G5 report the mean of the indicated characteristic in the bottom and top quintiles of the ML proxy score. G5-G1 reports the corresponding contrast followed by its 90 percent confidence interval in parentheses. * indicates that the median-aggregated 90 percent confidence interval excludes zero. Outcomes and characteristics are constructed such that a high value is better (except Preg. bf18). As such, a *positive* contrast (G5-G1) means that the treatment had a greater effect for AGYW in villages with *high* values of the characteristic at baseline. A *negative* contrast means treatment had a greater effect for AGYW in villages with *low* values of the characteristic at baseline. Estimates and confidence bounds are median-aggregated across 200 sample splits. See Appendix S6 for the GML methodology. Arm T4 combines T2 and T3 as if they were one treatment. ML algorithms are selected according to the goodness-of-fit of C6. Selection criteria are stated in Section S6.

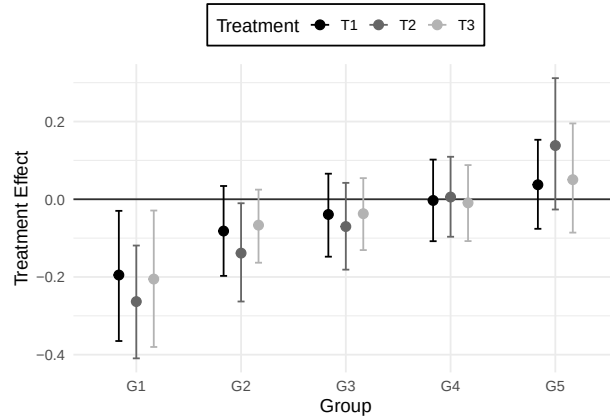
Figure S2: GATES – Economic and Social Outcomes



(a) Income-generating activities



(b) Mental-health index



(c) Emotional violence

Notes: Point estimates within each GATES group with 90 percent confidence regions, under the estimation framework of Section S6.

Village-level CATE decompositions

Each table regresses the village-level conditional average treatment effect for one dependent variable on the same seven standardized baseline structural predictors: columns 1–7 report the univariate regressions and column 8 the joint specification. Because the dependent variables are themselves estimates, the analytic standard errors ignore first-stage estimation error; the tables therefore carry no significance stars and the across-split sign-agreement rate is the stability metric used in the text.

Table S50: Predictors of village-level Take-up CATE (T3, strata FE)

	(1) Unrestricted	(2) Birthcontrols	(3) Econ. Infra.	(4) Security	(5) Male exog.	(6) Female exog.	(7) Overall exog.	(8) Joint
Unrestricted	0.033 (0.022)							0.040 (0.019)
Birthcontrols		0.059 (0.023)						0.061 (0.023)
Econ. Infra.			0.065 (0.024)					0.058 (0.024)
Security				-0.008 (0.028)				-0.016 (0.021)
Male exog.					0.026 (0.017)			0.009 (0.055)
Female exog.						0.033 (0.024)		0.011 (0.067)
Overall exog.							0.032 (0.017)	0.036 (0.095)
Strata FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N (villages)	84	84	84	84	84	84	84	84
R ²	0.396	0.430	0.453	0.380	0.392	0.398	0.398	0.551
F(7, 68)								3.78
p (joint F)								0.002

Notes: Dependent variable is the village-level CATE for Take-up under arm T3. Columns 1–7 regress on each of the seven village-level structural predictors entered alone (univariate); column 8 includes all seven jointly. All predictors are standardized (mean 0, SD 1); slopes are per-standard-deviation. Robust SEs throughout. Strata fixed effects: Yes.

CATE proxies are the GML *S*-statistic for each girl, aggregated as the median across 200 sample splits and collapsed to village means. Outcomes are coded so that a positive CATE is the desirable direction for every outcome except Preg. bf18.

No significance stars are reported: the dependent variables are estimated CATE scores and the analytic standard errors ignore first-stage estimation error. Sign stability across the 200 GML replications is reported in the split-uncertainty tables of this appendix.

Table S51: Predictors of village-level Never Pregnant CATE (T3, strata FE)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Unrestricted	Birthcontrols	Econ. Infra.	Security	Male exog.	Female exog.	Overall exog.	Joint
Unrestricted	-0.061 (0.016)							-0.064 (0.014)
Birthcontrols		0.008 (0.015)						-0.003 (0.014)
Econ. Infra.			0.037 (0.016)					0.040 (0.013)
Security				0.014 (0.022)				0.004 (0.016)
Male exog.					-0.023 (0.015)			-0.089 (0.031)
Female exog.						-0.010 (0.014)		-0.062 (0.039)
Overall exog.							-0.011 (0.015)	0.105 (0.054)
Strata FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N (villages)	84	84	84	84	84	84	84	84
R ²	0.259	0.108	0.172	0.113	0.132	0.110	0.112	0.417
F(7, 68)								6.23
p (joint F)								<0.001

Notes: Dependent variable is the village-level CATE for Never Pregnant under arm T3. Columns 1–7 regress on each of the seven village-level structural predictors entered alone (univariate); column 8 includes all seven jointly. All predictors are standardized (mean 0, SD 1); slopes are per-standard-deviation. Robust SEs throughout. Strata fixed effects: Yes.

CATE proxies are the GML *S*-statistic for each girl, aggregated as the median across 200 sample splits and collapsed to village means. Outcomes are coded so that a positive CATE is the desirable direction for every outcome except Preg. bf18.

No significance stars are reported: the dependent variables are estimated CATE scores and the analytic standard errors ignore first-stage estimation error. Sign stability across the 200 GML replications is reported in the split-uncertainty tables of this appendix.

Table S52: Predictors of village-level IGA CATE (T3, strata FE)

	(1) Unrestricted	(2) Birthcontrols	(3) Econ. Infra.	(4) Security	(5) Male exog.	(6) Female exog.	(7) Overall exog.	(8) Joint
Unrestricted	0.035 (0.062)							0.040 (0.064)
Birthcontrols		0.113 (0.084)						0.081 (0.067)
Econ. Infra.			-0.101 (0.050)					-0.125 (0.047)
Security				0.309 (0.069)				0.346 (0.068)
Male exog.					-0.142 (0.066)			0.077 (0.130)
Female exog.						-0.088 (0.079)		-0.037 (0.131)
Overall exog.							-0.146 (0.066)	-0.187 (0.198)
Strata FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N (villages)	84	84	84	84	84	84	84	84
R ²	0.321	0.340	0.340	0.459	0.362	0.335	0.363	0.559
F(7, 68)								7.56
p (joint F)								<0.001

Notes: Dependent variable is the village-level CATE for IGA under arm T3. Columns 1–7 regress on each of the seven village-level structural predictors entered alone (univariate); column 8 includes all seven jointly. All predictors are standardized (mean 0, SD 1); slopes are per-standard-deviation. Robust SEs throughout. Strata fixed effects: Yes.

CATE proxies are the GML *S*-statistic for each girl, aggregated as the median across 200 sample splits and collapsed to village means. Outcomes are coded so that a positive CATE is the desirable direction for every outcome except Preg. bf18.

No significance stars are reported: the dependent variables are estimated CATE scores and the analytic standard errors ignore first-stage estimation error. Sign stability across the 200 GML replications is reported in the split-uncertainty tables of this appendix.

Table S53: Structural predictors of village-level family-index CATEs, by arm

	Mar. fert. Index	Econ. Index	Kdge Index	Pref. Index	Wl. being Index
<i>Panel A: T1</i>					
Unrestricted (norms)	-0.068 (0.016)	0.034 (0.062)	0.106 (0.068)	0.011 (0.046)	-0.109 (0.063)
Birth-control infr.	-0.032 (0.015)	0.084 (0.062)	0.095 (0.072)	0.018 (0.050)	0.046 (0.066)
Econ. infrastructure	0.041 (0.013)	-0.030 (0.043)	-0.121 (0.046)	-0.080 (0.048)	0.115 (0.055)
Security	-0.002 (0.022)	0.061 (0.057)	0.046 (0.071)	0.016 (0.073)	0.028 (0.072)
Male exogamy	0.002 (0.016)	-0.023 (0.051)	0.053 (0.053)	-0.011 (0.051)	0.077 (0.052)
Female exogamy	0.023 (0.015)	-0.134 (0.048)	-0.264 (0.070)	-0.079 (0.052)	0.096 (0.052)
Overall exogamy	0.022 (0.015)	-0.102 (0.048)	-0.108 (0.060)	-0.037 (0.049)	0.112 (0.052)
Joint F (7 pred.)	6.42	1.73	8.02	1.27	2.21
p (joint F)	0.000	0.120	0.000	0.283	0.046
N (villages)	76	76	76	76	76
<i>Panel B: T2</i>					
Unrestricted (norms)	-0.102 (0.019)	-0.008 (0.045)	0.015 (0.055)	0.054 (0.051)	-0.040 (0.074)
Birth-control infr.	-0.050 (0.021)	0.219 (0.051)	0.081 (0.052)	0.080 (0.055)	-0.060 (0.067)
Econ. infrastructure	0.041 (0.016)	-0.021 (0.045)	-0.129 (0.054)	-0.083 (0.040)	0.090 (0.057)
Security	0.005 (0.025)	0.049 (0.071)	0.020 (0.068)	0.089 (0.077)	-0.123 (0.105)
Male exogamy	0.141 (0.014)	-0.134 (0.047)	-0.105 (0.043)	-0.120 (0.045)	-0.079 (0.055)
Female exogamy	0.063 (0.022)	-0.151 (0.049)	-0.210 (0.045)	-0.126 (0.049)	-0.163 (0.067)
Overall exogamy	0.128 (0.013)	-0.166 (0.043)	-0.172 (0.044)	-0.131 (0.045)	-0.134 (0.058)
Joint F (7 pred.)	66.96	7.92	5.03	5.05	2.13
p (joint F)	0.000	0.000	0.000	0.000	0.053
N (villages)	78	78	78	78	78
<i>Panel C: T3</i>					
Unrestricted (norms)	0.004 (0.025)	0.022 (0.052)	0.152 (0.061)	0.152 (0.053)	0.071 (0.070)
Birth-control infr.	0.002 (0.020)	0.118 (0.059)	0.022 (0.051)	-0.023 (0.060)	0.122 (0.066)
Econ. infrastructure	0.016 (0.015)	-0.027 (0.041)	-0.005 (0.060)	-0.020 (0.045)	0.094 (0.065)
Security	0.099 (0.014)	0.032 (0.068)	-0.139 (0.065)	0.040 (0.074)	-0.126 (0.068)
Male exogamy	-0.034 (0.018)	-0.011 (0.057)	-0.007 (0.054)	-0.136 (0.049)	0.016 (0.058)
Female exogamy	0.008 (0.024)	-0.069 (0.052)	-0.184 (0.049)	0.003 (0.054)	-0.102 (0.077)
Overall exogamy	-0.004 (0.023)	-0.056 (0.058)	-0.123 (0.046)	-0.071 (0.048)	-0.023 (0.063)
Joint F (7 pred.)	8.12	1.81	4.96	3.32	2.73
p (joint F)	0.000	0.099	0.000	0.004	0.015
N (villages)	84	84	84	84	84
<i>Panel D: T4</i>					
Unrestricted (norms)	-0.028 (0.017)	-0.015 (0.041)	0.034 (0.050)	0.057 (0.049)	0.027 (0.062)
Birth-control infr.	-0.020 (0.019)	0.191 (0.044)	0.059 (0.040)	0.031 (0.047)	0.008 (0.059)
Econ. infrastructure	0.018 (0.013)	-0.042 (0.037)	-0.063 (0.050)	-0.041 (0.038)	0.105 (0.059)
Security	0.029 (0.011)	0.036 (0.048)	-0.105 (0.046)	0.052 (0.055)	-0.103 (0.063)
Male exogamy	0.041 (0.014)	-0.039 (0.046)	-0.039 (0.041)	-0.118 (0.036)	0.014 (0.048)
Female exogamy	0.023 (0.014)	-0.088 (0.041)	-0.186 (0.040)	-0.068 (0.042)	-0.141 (0.057)
Overall exogamy	0.045 (0.014)	-0.075 (0.044)	-0.136 (0.039)	-0.104 (0.038)	-0.062 (0.053)
Joint F (7 pred.)	3.58	4.67	5.43	2.28	2.04
p (joint F)	0.002	0.000	0.000	0.033	0.057
N (villages)	124	124	124	124	124

Notes: Cells are slopes from separate univariate OLS regressions of the column's village-level family-index CATE on the row's standardized (mean 0, SD 1) village predictor, with strata fixed effects; robust SEs in parentheses; slopes are per SD of the predictor. Joint F, its p-value and N are from the regression including all seven predictors jointly (strata FE, robust SEs). The family indices aggregate the individual outcomes by domain and were not pre-specified; the all-domain super-index is omitted because it pools domains whose CATEs align with take-up in opposite directions. Positive CATE = beneficial direction. CATE proxies are the GML *S*-statistic, median across 200 sample splits. No significance stars are reported: the dependent variables are estimated CATE scores and the analytic standard errors ignore first-stage

Table S54: Participation-Barrier Test — Robustness to Village Moderators (T3)

Outcome CATE	Headline		+ Moderators	
	Slope	p-value	Slope	p-value
Never Married	-0.250 (0.425)	0.558	0.196 (0.473)	0.679
Age at Mar.	-0.060 (0.068)	0.383	-0.022 (0.066)	0.738
Mar. af18	0.981 (0.551)	0.079	-0.177 (0.691)	0.799
Never Pregnant	-0.360 (0.126)	0.005	-0.512 (0.177)	0.005
Curr. Not Preg.	-1.832 (1.253)	0.148	-1.741 (1.151)	0.135
Preg. bf18	-0.046 (0.420)	0.914	0.164 (0.500)	0.743
No Emo. Viol.	0.328 (0.248)	0.191	0.140 (0.374)	0.709
IGA	-0.035 (0.032)	0.279	-0.018 (0.031)	0.568
Mental Health	0.037 (0.038)	0.325	0.017 (0.032)	0.601
Voc. Training	0.853 (0.122)	0.000	0.816 (0.103)	0.000
Prof. Training	1.323 (0.235)	0.000	1.106 (0.245)	0.000
Finance	0.171 (0.040)	0.000	0.161 (0.035)	0.000
Gender Att.	-0.021 (0.059)	0.724	-0.019 (0.058)	0.740
Aspirations	0.012 (0.052)	0.821	0.049 (0.051)	0.340
SRH Knowledge	-0.028 (0.046)	0.547	-0.039 (0.053)	0.461
SRH Behavior	0.048 (0.042)	0.252	-0.009 (0.048)	0.856
MHM	-0.020 (0.056)	0.715	0.024 (0.054)	0.666
Decision-making	0.048 (0.038)	0.215	0.069 (0.035)	0.052
Decision (couple)	-0.002 (0.040)	0.964	0.028 (0.036)	0.445
Men's Score	0.064 (0.033)	0.058	0.051 (0.032)	0.112
SES	0.018 (0.032)	0.579	-0.009 (0.026)	0.737
EM Negative	0.074 (0.108)	0.493	-0.013 (0.110)	0.905
EM Neg. (mult.)	0.030 (0.031)	0.341	-0.004 (0.031)	0.899
EM Positive	-0.137 (0.117)	0.245	-0.117 (0.110)	0.293
EM Pos.(mult.)	0.018 (0.046)	0.701	-0.025 (0.040)	0.538
Men's Score (mult.)	0.058 (0.058)	0.000	0.010 (0.010)	0.000

Table S55: Within-Village Individual Participation-Barrier Test (T3)

Outcome CATE	Slope	p-value	N
Never Married	0.037 (0.050)	0.457	2596
Age at Mar.	0.037 (0.005)	0.000	1993
Mar. af18	0.161 (0.039)	0.000	1949
Never Pregnant	-0.241 (0.029)	0.000	2596
Curr. Not Preg.	-0.687 (0.181)	0.000	2439
Preg. bf18	0.203 (0.040)	0.000	2154
No Emo. Viol.	0.110 (0.073)	0.132	2372
IGA	-0.011 (0.014)	0.433	2595
Mental Health	0.043 (0.017)	0.012	2596
Voc. Training	0.629 (0.039)	0.000	2596
Prof. Training	0.335 (0.050)	0.000	2577
Finance	0.078 (0.017)	0.000	2596
Gender Att.	-0.020 (0.019)	0.294	2596
Aspirations	0.003 (0.015)	0.845	2596
SRH Knowledge	-0.092 (0.014)	0.000	2177
SRH Behavior	-0.058 (0.012)	0.000	2177
MHM	-0.036 (0.015)	0.018	2197
Decision-making	0.069 (0.012)	0.000	2596
Decision (couple)	0.069 (0.010)	0.000	1662
Men's Score	0.042 (0.010)	0.000	2582
SES	-0.010 (0.012)	0.392	2596
EM Negative	0.003 (0.022)	0.905	2596
EM Neg. (mult.)	-0.092 (0.011)	0.000	2596
EM Positive	0.072 (0.050)	0.152	2596
EM Pos. (mult.)	0.047 (0.023)	0.046	2596
Mar. fert. Index	-0.073 (0.008)	0.000	2596

Assignment interactions with randomization inference

Table S56: Assignment $\times$ village-structure interactions with randomization inference (T3, all outcomes)

Village predictor X_v	Take-up	Never Pregnant	IGA	Age at Mar.	Mar. af18	Curr. Not Preg.	Preg. bf18	No Emo. Viol.	Mental Health
Unrestricted	0.069 [0.162]	-0.081 [0.110]	0.127 [0.375]	0.182 [0.269]	0.023 [0.515]	-0.003 [0.878]	-0.014 [0.722]	-0.014 [0.632]	0.103 [0.529]
Birthcontrols	0.106** [0.016]	0.007 [0.896]	0.252 [0.156]	0.009 [0.964]	0.020 [0.619]	0.006 [0.767]	0.029 [0.494]	-0.003 [0.915]	0.200 [0.308]
Econ. Infra.	0.103** [0.043]	0.105** [0.045]	-0.242* [0.086]	-0.052 [0.759]	-0.044 [0.389]	0.007 [0.705]	-0.034 [0.179]	0.045 [0.161]	-0.050 [0.713]
Security	-0.031 [0.532]	0.067 [0.316]	0.484*** [0.001]	0.091 [0.630]	0.028 [0.513]	0.003 [0.895]	-0.053 [0.207]	-0.040 [0.255]	-0.064 [0.772]
Male exog.	0.070 [0.116]	-0.043 [0.298]	-0.271 [0.141]	0.142 [0.332]	0.003 [0.928]	-0.010 [0.600]	0.025 [0.346]	0.034 [0.189]	0.023 [0.889]
Female exog.	0.021 [0.658]	-0.031 [0.522]	-0.112 [0.568]	0.003 [0.981]	0.033 [0.382]	-0.017 [0.302]	-0.026 [0.386]	-0.029 [0.564]	-0.023 [0.869]
Overall exog.	0.055 [0.200]	-0.009 [0.871]	-0.241 [0.232]	0.144 [0.340]	0.036 [0.371]	-0.007 [0.693]	-0.016 [0.614]	0.010 [0.736]	0.051 [0.733]

Notes: Coefficients γ on $T3 \times X_v$ from the specification of the main-paper assignment-interaction table; randomization-inference p -values in brackets (10000 within-strata permutations of the four-category assignment across villages, arm counts fixed, $|t|$ criterion). Stars are on RI p -values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. These 63 interactions form the exploratory T3 family; Benjamini–Hochberg q -values across the family are reported in the replication package. The exploratory focal subset is reported with familywise correction in the main-paper focal-interaction table.

No focal interaction survives familywise correction. Security–IGA is distinct from the two-moderator participation hypothesis. The full T3 grid makes that selection visible; it does not make the focal family pre-specified.

Follow-up response by moderator tercile

Table S57 first reports the linear assignment-by-moderator interaction on follow-up response. Because a linear interaction can miss a gap concentrated in a single tercile, the tercile-parameterized version follows.

Table S57: Assignment $\times$ village-structure interactions on follow-up response

Village predictor X_v	T1		T2		T3	
	γ	p	γ	p	γ	p
Unrestricted	0.129**	0.016	0.088**	0.024	0.020	0.535
Birthcontrols	-0.025	0.608	0.021	0.599	-0.009	0.785
Econ. Infra.	-0.049	0.498	-0.034	0.504	-0.053	0.201
Security	0.097*	0.086	0.060	0.256	0.015	0.640
Male exog.	0.011	0.832	-0.002	0.966	0.005	0.892
Female exog.	0.046	0.388	0.054	0.245	0.027	0.472
Overall exog.	0.030	0.576	0.030	0.533	0.006	0.865

Notes: Coefficient γ on $Tk \times X_v$ from $\text{respond}_i = \alpha + \beta Tk + \gamma (Tk \times X_v) + \delta X_v + \text{strata FE}$ on the full baseline roster (Tk-union-control), where $\text{respond} = 1$ if the girl is observed at follow-up. X_v standardized as in the design-based interaction table of the paper's Appendix C. Village-clustered standard errors. A significant γ indicates that assignment shifts follow-up response differentially across village types—the selection channel that could bias assignment $\times$ moderator outcome interactions. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table S58 reports the raw follow-up response rates by moderator tercile and arm, before any strata adjustment.

Table S58: Sample flow by treatment arm and moderator tercile

Moderator	Tercile	Arm	Baseline N	Follow-up N	Response rate
Unrestricted	1 (lowest)	Control	564	346	0.613
Unrestricted	1 (lowest)	T3	688	515	0.749
Unrestricted	2	Control	468	340	0.726
Unrestricted	2	T3	807	570	0.706
Unrestricted	3 (highest)	Control	672	391	0.582
Unrestricted	3 (highest)	T3	693	434	0.626
Econ. Infra.	1 (lowest)	Control	493	234	0.475
Econ. Infra.	1 (lowest)	T3	826	536	0.649
Econ. Infra.	2	Control	632	423	0.669
Econ. Infra.	2	T3	714	500	0.700
Econ. Infra.	3 (highest)	Control	579	420	0.725
Econ. Infra.	3 (highest)	T3	648	483	0.745

Notes: Baseline and follow-up counts of adolescent respondents by treatment arm and overall cross-village tercile of the moderator (terciles computed across all study villages). Response rate = follow-up N / baseline N . T1 and T2 flows are reported in the replication package.

Table S59: Assignment effects on follow-up response, by moderator tercile

	T1		T2		T3	
	$\hat{\beta}_t$	p_{RI}	$\hat{\beta}_t$	p_{RI}	$\hat{\beta}_t$	p_{RI}
<i>Unrestricted terciles</i>						
Tercile 1 (lowest)	-0.122	0.292	-0.088	0.383	0.031	0.740
Tercile 2	-0.180**	0.082	-0.086	0.228	0.010	0.892
Tercile 3 (highest)	0.171**	0.038	0.139*	0.084	0.104	0.146
Equality: analytic / RI p	0.015	0.018	0.059	0.122	0.575	0.633
<i>Econ. Infra. terciles</i>						
Tercile 1 (lowest)	0.060	0.685	0.100	0.364	0.208**	0.086
Tercile 2	-0.128	0.182	-0.040	0.637	-0.028	0.577
Tercile 3 (highest)	-0.052	0.487	-0.035	0.483	0.023	0.678
Equality: analytic / RI p	0.486	0.589	0.429	0.519	0.102	0.174

Notes: Coefficients $\hat{\beta}_t$ on $Tk \times 1[\text{tercile} = t]$ from $\text{respond}_i = \sum_t \beta_t (Tk \times 1[\text{terc}_V = t]) + \text{tercile indicators} + \text{strata FE}$, estimated on the full baseline roster (Tk-union-control), where $\text{respond} = 1$ if the girl is observed at follow-up; each $\hat{\beta}_t$ is the strata-adjusted arm–control difference in follow-up response within tercile t of the baseline village moderator. Terciles are computed across all study villages. Stars reflect analytic p -values with village-clustered standard errors; p_{RI} is the two-sided randomization-inference p -value from 1,000 permutations of the four-category arm label across villages within strata (studentized statistic; plus-one convention). The equality row reports the Wald test that the three tercile coefficients are equal (analytic p and its RI counterpart based on the F statistic). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

T3’s response gap is largest in the weakest-infrastructure tercile (0.208; RI $p = 0.086$), the margin contributing to the income pattern. The T3 norms-tercile equality test is not significant. Lack of detectable differential selection does not make common attrition ignorable.

Selection sensitivity of focal interactions

Table S60: Attrition sensitivity of the exploratory focal interaction family (T3)

Predictor X_v	Outcome	γ (baseline)	γ (IPW)	γ (high-retention)	Flip threshold k^*
Unrestricted	Take-up	0.069	0.068	0.089	—
Unrestricted	Never Pregnant	-0.081	-0.097	-0.082	0.4
Unrestricted	IGA	0.127	0.146	0.026	—
Econ. Infra.	Take-up	0.103	0.102	0.089	—
Econ. Infra.	Never Pregnant	0.105	0.084	0.104	—
Econ. Infra.	IGA	-0.242	-0.223	0.093	0.4

Notes: γ (baseline) reproduces the main-paper focal-interaction table. γ (IPW) reweights respondents by the inverse of a logit response model estimated on the full baseline roster (age, age squared, baseline schooling, baseline ever-pregnant, the seven village moderators, assignment, assignment $\times$ {Unrestricted, Econ. Infra.}, strata dummies; weights trimmed at the 99th percentile). γ (high-retention) restricts to villages with follow-up response rates of at least 65 percent. k^* is the adversarial-imputation flip threshold: attritors' outcomes are imputed at their arm $\times$ above/below-median- X_v cell mean shifted by k within-cell standard deviations in the directions chosen to overturn the observed interaction; k^* is the smallest $k \in \{0, 0.1, \dots, 1.0\}$ at which the interaction changes sign (> 1.0 if no flip occurs). Village-clustered standard errors throughout.

Infrastructure-IGA changes from -0.242 to $+0.093$ in villages with at least 65% retention. Norms-pregnancy remains negative. Neither establishes a joint barrier mechanism: that requires a reliable take-up relationship, and retention restrictions can change the represented population.

Implementation sensitivity of focal interactions

Table S61: Implementation fidelity and the exploratory focal interaction family (T3)

Predictor X_v	Outcome	Baseline	Excl. suspended	Excl. disruption prov.	T3 $\times$ disruption ctrl.
Unrestricted	Take-up	0.069	0.064	-0.034	-0.019
Unrestricted	Never Pregnant	-0.081*	-0.094**	-0.076	-0.077
Unrestricted	IGA	0.127	0.125	0.254	0.121
Econ. Infra.	Take-up	0.103**	0.102**	0.113*	0.084**
Econ. Infra.	Never Pregnant	0.105***	0.104***	0.105	0.112***
Econ. Infra.	IGA	-0.242**	-0.242**	-0.157	-0.259**

Notes: Coefficient γ on $T3 \times X_v$ from the specification of the main-paper assignment-interaction table under implementation-fidelity checks. *Excl. suspended* drops the four villages whose safe spaces were suspended from October 2019 (Dadounga and Santiari in Gnagna; Boumontchangou and Sampieri in Tapoa; ASMADE final implementation report). *Excl. disruption prov.* drops all villages in provinces with documented implementation disruption (Gnagna and Tapoa suspensions; Sourou delocalizations; unvisited security-risk provinces where present in the sample). *T3 $\times$ disruption ctrl.* retains the full sample and adds a disruption-province indicator and its interaction with assignment. The thirteen Est-region safe spaces displaced within zone are not individually identified in program records; disruption-province exclusion is the feasible bound on their influence, and this residual limitation is stated in the text. Stars use analytic village-clustered p -values: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Norms—take-up is absorbed or reversed after disruption controls or exclusions. The pregnancy outcome gradient remains negative, but a barrier account requires both legs. Income sensitivity to selection and the competing substitution interpretation preclude an infrastructure recommendation based on these interactions alone.

Exogamy: a partial test of the framework

The prediction requires positive male-exogamy contrasts on marriage delay and negative female-exogamy contrasts on IGA. A significant difference alone is insufficient. An intersection–union reading supports both legs only for T2’s never-married/IGA pair ($\max\{0.045, 0.015\} = 0.045$). Elsewhere evidence is one-legged. These post-registration tests offer partial support, not model validation.

Table S62: Exogamy Falsification Test: Sign Reversal Across Outcome Families (the main-paper exogamy proposition)

	T1: Safe Spaces	T2: SS + Training	T3: Full Package
<i>Panel A: Male exogamy (σ_v^g) on marriage-delay outcomes</i>			
	−0.021	+0.309**	−0.045
Never Married	[−0.076, +0.039] $p_H = 1.000$	[+0.220, +0.389] $p_H = 0.045$	[−0.106, +0.021] $p_H = 1.000$
	+0.088	−0.020	−0.049
Age at Mar.	[+0.003, +0.178] $p_H = 0.800$	[−0.081, +0.080] $p_H = 1.000$	[−0.133, +0.040] $p_H = 1.000$
	+0.076	+0.077	−0.001
Mar. af18	[−0.006, +0.177] $p_H = 0.980$	[−0.049, +0.180] $p_H = 1.000$	[−0.082, +0.103] $p_H = 1.000$
<i>Panel B: Female exogamy (σ_v^b) on income-generation outcomes</i>			
	−0.156**	−0.187**	−0.110**
IGA	[−0.226, −0.094] $p_H = 0.015$	[−0.268, −0.089] $p_H = 0.015$	[−0.174, −0.047] $p_H = 0.020$
<i>Panel C: Cross-family sign-reversal test</i>			
	+0.202**	+0.312**	+0.081
$\bar{\delta}_{\sigma_g^{\text{delay}}} - \delta_{\sigma_b^{\text{IGA}}}$	[+0.135, +0.292] $p_H = 0.015$	[+0.206, +0.407] $p_H = 0.015$	[+0.001, +0.161] $p_H = 0.100$
Notes: Each cell in Panels A and B reports the CLAN δ estimate (median across replications), its 90% empirical CI from the rep distribution, and the Holm-adjusted two-sided p -value. $\delta(X) = E[X i \in G_5] - E[X i \in G_1]$, where G_1 and G_5 are the bottom and top quintiles of the per-observation S -statistic within a replication. Best ML learner per (arm, outcome) is selected by the $\hat{\Lambda}$ goodness-of-fit criterion. Holm correction is applied within each panel. Panel C reports the cross-family contrast $\bar{\delta}_{\sigma_g^{\text{delay}}} - \delta_{\sigma_b^{\text{IGA}}}$, with the marriage-delay average taken across the three Panel A outcomes; rejection of H_0 : contrast = 0 establishes a difference, not the conjunction of a positive delay leg and negative IGA leg. σ_v^g = male exogamy, σ_v^b = female exogamy, both at the village level (constant within cluster). σ_v^g + outside-option theory predicts $\delta^{\text{delay}} > 0$ and $\delta^{\text{IGA}} < 0$. * $p_H < 0.10$, ** $p_H < 0.05$, *** $p_H < 0.01$.			